PHYSICS

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. If electron of hydrogen atom is replaced by another particle of same charge but of double mass then

(A) Bohr's radius will increase

(B) Ionisation energy of atom will be doubled

(C) Speed of new particle in given state will be lesser than electrons speed in same orbit

(D) Gap between energy levels will now be doubled

2. A metal spherical shell of mass m is filled completely with nonviscous water of mass m/2 and is rolled down an inclined plane of height h. The total kinetic energy of shell plus water and the velocity of the centre of the shell when it reaches the bottom of the plane is K and v respectively. If water in the shell is freezed and then it is rolled down, the corresponding values of kinetic energy and velocity are K' and v' respectively. Then :

(A) $K = \frac{3}{2}$ mgh (B) $K' = \frac{3}{2}$ mgh (C) $v = v'$ (D) $v > v'$

3. A spherical planet of radius R has spherically symmetrical distribution of mass density, varying as square of the distance from the centre, from zero at centre to maximum value ρ_0 at its surface.

(A) The value of escape velocity of a mass m at the surface of planet is $\sqrt{\frac{4\pi G\rho_0 R^2}{5}}$.

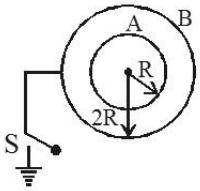
(B) The value of acceleration due to gravity 'g' varies inside the planet as cube of the distance from centre.

(C) The value of escape velocity from surface is same as the escape velocity from another planet of same total mass & radius but having uniform mass density.

(D) The energy required to impart escape velocity to particles of masses 'm' & '2m', at the surface of planet, will be in ratio 1:2.

Space for Rough Work

4. Two concentric conducting spherical shells of radii R and 2R having charges q_A and q_B and potential 2v and 3v/2 respectively. Now shell B is earthed by closing switch S. Let charges on spherical shells A & B become q'_A & q'_B respectively, then



(A) $q_A/q_B = \frac{1}{2}$ (magnitude).

(B) $q'_A/q'_B = 1$ (magnitude).

(C) potential of A after earthing becomes $\frac{3v}{2}$.

(D) potential difference between A and B $(v_A - v_B)$ after earthing becomes $\frac{v}{2}$.

5. The magnetic flux (ϕ) linked with a coil depends on time t as $\phi = at^n$, where a is a constant. The induced e.m.f. in the coil is e :

(A) If $0 < n < 1$, $e = 0$

(B) If $0 < n < 1$, $e \neq 0$ and $|e|$ decreases with time

(C) If $n = 1$, e is constant

(D) If $n > 1$, $|e|$ increases with time

Space for Rough Work

6. Sound waves from different source are interfering and pressure at some point in space is given as –
$P = P_0 \sin(200\pi t + \pi/2) \cos(3\pi t + \pi/3)$

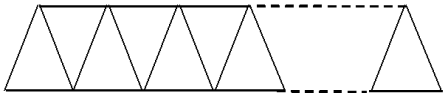
(A) The frequency of one source is above 100Hz and other source is below 100Hz.

(B) Beat frequency is 3 Hz

(C) Beat frequency is 200 Hz

(D) Beat frequency can be 3 Hz as well as 200 Hz

7. n number of identical equilateral prism are kept in contact as shown in figure. If deviation through a single prism is δ. Then - (n, m are integers)



(A) If $n = 2m$, deviation through n prisms is zero.

(B) If $n = 2m + 1$, deviation through system of n prisms is δ.

(C) If $n = 2m$, deviation through system of n prisms is δ.

(D) If $n = 2m + 1$, deviation through system of n prisms is zero.

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

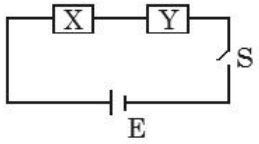
- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–1	In all other cases.

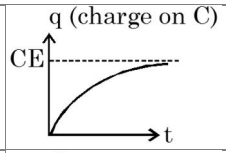
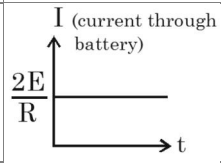
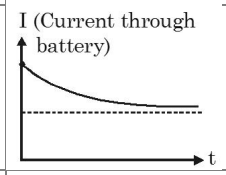
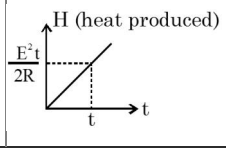
Space for Rough Work

Paragraph for Questions 8 - 10

Following is a circuit containing an ideal battery of emf E in which the elements X & Y are unknown & may be a capacitor C, a resistor R, a battery of emf E, or may be their combination as mentioned in each case. The connecting wires are perfectly conducting.



The switch S is initially open & it is closed at t = 0.

Column-1		Column-2		Column-3	
(I)	Element X is R Element Y is C	(i)	Entire energy (non–zero) supplied by batteries is dissipated as heat	(P)	
(II)	Both elements X & Y are R each	(ii)	Total charge drawn from the battery(s) after long time is CE/2	(Q)	
(III)	Element X is R, element Y is a parallel combination of R and C	(iii)	Total heat produced in circuit is less than the energy supplied by the battery(s)	(R)	
(IV)	Element X is R element Y is battery E joined with same polarity	(iv)	I = 0 just after S is closed as well as after long time	(S)	

1. Of the given 4 situations, which one has maximum value of current just after closing the switch ?

(A) I, iii, P (B) IV, i, Q (C) II, iii, P (D) III, i, S

Space for Rough Work

2. In which case the charge drawn from the battery(s) during t = 0 to t = RC is the least ?

(A) II, i, S (B) I, iii, P (C) III, ii, Q (D) IV, i, Q

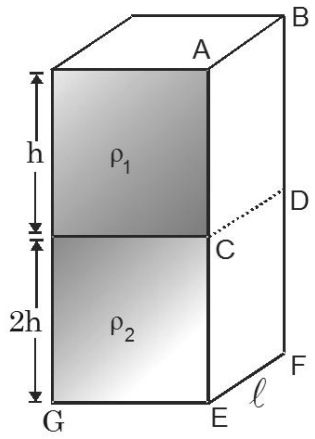
3. In which case the current through battery(s) has different non zero values just after closing the switch & after a long time ?

(A) II, i, S (B) I, iv, Q (C) IV, i, Q (D) III, iii, R

Space for Rough Work

Paragraph for Questions 11 - 13

A cuboid is filled with liquid of density ρ_2 upto height 2h & with liquid of density ρ_1, upto height h as shown in the figure and assume pressure at top to be zero.



Column-1		Column-2		Column-3	
(I)	Force on face ABCD due to liquid of density ρ_1	(i)	zero	(P)	Effectively acts at centre of that face
(II)	Force on face ABCD due to liquid of density ρ_2	(ii)	$\frac{\rho_1 g h^2 \ell}{2}$	(Q)	Effectively acts at $\frac{h}{3}$ above the bottom of that face
(III)	Magnitude of increase in force on face CDEF due to filling liquid of density ρ_1	(iii)	$2\rho_1 gh^2\ell$	(R)	Effectively acts at $\frac{4h}{3}$ below the top of that face
(IV)	Force on face CDEF if liquid of density ρ_2 only is present	(iv)	$2\rho_2 gh^2\ell$	(S)	Cannot produce a net torque about bottom edge of that face

4. Which force has the second highest magnitude (given $\rho_1 < \rho_2$) ?

(A) I,ii, Q (B) IV, i,Q (C) II,i, S (D) III,iii, P

Space for Rough Work

5. Which force produces minimum (Non zero) torque about side E F (given $\rho_1 = \rho_2$) ?

(A) I,ii, Q (B) II,i, S (C) III,iii, P (D) IV, iii, S

6. Which force is equal to the total weight of liquid ρ_2 [Given GE = h]?

(A) I, iii, P (B) II,i, S (C) III,iii, P (D) IV, iv, R

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

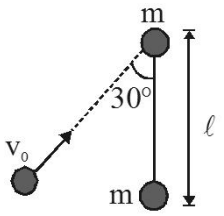
- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
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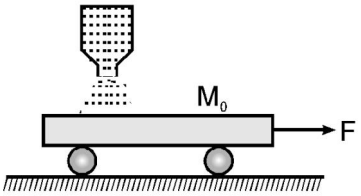
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1. A small putty of mass m = 1 kg and speed v_0 = 4 m/s, strikes a dumbbell placed on a smooth table as shown in figure. Putty is moving in the plane of the dumbbell. The putty sticks to dumbbell ball which is to be treated as particle. How much thermal energy is produced (in J). Consider rod of dumbbell to be massless.

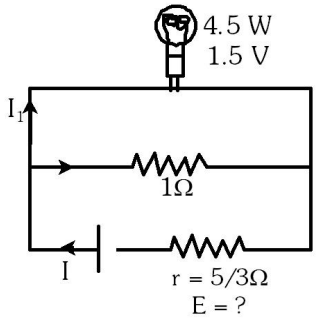


Space for Rough Work

2. In the figure shown a cart moves on a smooth horizontal surface with a constant velocity v acted upon by an external constant force of magnitude F. Sand falls on to the cart with negligible velocity at constant rate μ kg/s and sticks to the cart. Find N if the power supplied by external agent applying the force F is μv^N.

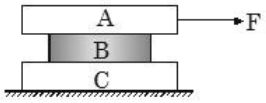


3. A torch bulb rated at 4.5 W, 1.5 V is connected as shown in the figure. To make the bulb glow at full intensity, what should be the emf (in volt) of the cell?

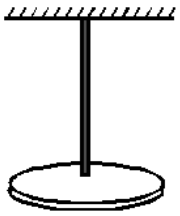


Space for Rough Work

4. Given $m_A = 30$ kg, $m_B = 10$ kg, $m_C = 20$ kg. Between A and B; $\mu_1 = 0.3$, between B and C; $\mu_2 = 0.2$ and between C and ground $\mu_3 = 0.1$. The least horizontal force which can start the motion of any part of the system of three blocks resting upon one another as shown in figure is F. Write the value of F/12 ($g = 10$ m/s^2)



5. A circular disc suspended from a torsional wire, oscillates with a time period T when twisted by a small angle θ from equilibrium and released.



If another disc of same material but twice the radius and one fourth thickness is made to oscillate (alone) with same wire, its time period is found to be nT. Write the value of n.

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. Which of the following statement(s) is/are correct ?

(A) Glucose & Fructose are optical isomers.

(B) Fructose can be oxidised by tollens reagent.

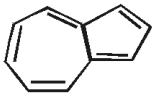
(C) Monosaccharides containing six membered ring can be called pyranose sugar.

(D) Sucrose is a disaccharide and it contains two glycosidic linkages.

2. Which of the following statement(s) is/are incorrect ?

(A) Bakelite is a polyamide

(B) All Biodegradable polymers are natural polymers

(C) Azulene  has zero dipole moment.

(D) Sulphonation of C_6H_6, C_6D_6 & C_6T_6 by Electrophilic aromatic substitution mechanism proceed at same rate.

3. Select incorrect statement(s) :

(A) pka value of $B(OH)_3$(aq.) increases on adding glycerol.

(B) aqueous solution of $Ca(OH)_2$ can not be used for removal of temporary hardness of water.

(C) Crystals of borax on heating does not produce any neutral oxide.

(D) Two layers of solid orthoboric acid are connected together by hydrogen bonds.

Space for Rough Work

4. For thermal decomposition of $NH_4NO_3(s)$:
The correct information(s) is/are -

$$NH_4NO_3(s) \xrightarrow{\Delta_r H} N_2O(g) + 2H_2O(g) \quad (\text{at T K \& P atm})$$

(A) Heat of reaction is positive

(B) Entropy change of the reaction must be $\frac{\Delta_r H}{T}$

(C) Heat of reaction at constant volume is equal to $(\Delta_r H - 3\ RT)$

(D) $\Delta_r G$ of this reaction may be negative at high temperature.

5. Select the correct statement(s) regarding following galvanic cell taken at temperature TK.
$Pt \mid H_2(g, 1\ bar) \mid HCl\ (10^{-3}\ M) \mid AgCl(s) \mid Ag(s)$
Given that : $\frac{2.303RT}{F} = 0.06 volt$
& $E^{\circ}_{Ag|AgCl|Cl^-} = -\ 0.18 volt$

(A) Cell potential of this cell is 0.54 volt.

(B) During discharging, $H_2(g)$ converts into H^+ (aq.)

(C) $H_2(g)$ is obtained at cathode during working of this galvanic cell.

(D) $E^{\circ}_{Ag^+|Ag}$ is equal to 0.78 volt if K_{sp} of AgCl is 1×10^{-10}.

Space for Rough Work

6. Select the correct option(s) about valence Bond Theory regarding co-ordination compounds:

(A) It does not explain visible spectra of complexes.

(B) It is not able to explain diamagnetic behaviour of $[Co(H_2O)_6]Cl_3$ even Co^{3+} is paramagnetic.

(C) It is unable to differentiate between weak and strong field ligands.

(D) It makes exact predictions regarding the tetrahedral and square planar structures of 4-coordinate complexes.

7. Which of the following step(s) is/are involved in extraction of iron from haematite ore ?

(A) Gravity separation

(B) Reduction by CO

(C) Self-reduction

(D) Slag formation

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–1	In all other cases.

Space for Rough Work

Paragraph for Questions 8 - 10

For given solution in the questions molarity can be treated as molality and solute(s) is/are nonvolatile in nature.

Given that : K_a of $CH_3COOH = 10^{-5}$; K_f = Molal depression constant

K_a of $HCOOH = 10^{-4}$; K_b = Molal elevation constant

Column - I (Colligative properties)		Column - II (One of the product formed during electrolysis at Pt - electrode)		Column - III (Nature of aqueous solution at 25°C)	
(I)	$\Delta T_b \simeq 0.3\ K_b$	(A)	Metal becomes deposited at cathode	(P)	pH = 8.5
(II)	$\pi \simeq 0.4\ RT$	(B)	H_2 gas at cathode	(Q)	pH = 4.7
(III)	$\Delta T_f \simeq 0.2\ K_f$	(C)	H_2 gas at anode	(R)	pH $\approx$ 7
(IV)	$\pi \simeq 0.3\ RT$	(D)	O_2 gas at anode	(S)	Weakly basic solution

1. Which of the following option is correctly matched for 0.1 M HCOOK(aq.)

(A) (I),(B),(S) (B) (IV), (B), (S) (C) (III),(D), (Q) (D) (III),(C),(P)

2. Select incorrect match for 0.1 m $Ca(NO_3)_2$ aqueous solution.

(A) (I),(B),(R) (B) (III),(A),(P) (C) (IV),(D),(R) (D) (I),(D),(R)

3. Select the correct match for aq. solution having 0.1 M CH_3COONa and 0.2 M CH_3COOH.

(A) (II),(B),(Q) (B) (II),(A),(Q) (C) (IV),(A),(S) (D) (I),(C),(R)

Space for Rough Work

Paragraph for Questions 11 - 13

Column - I (Name reaction or test)		Column - II (Related Reagent or intermediate)		Column - III (Miscellaneous)	
(I)	Hoffman bromamide reaction	(i)	Carbene	(P)	Reaction involves upgradation of chain
(II)	Reimer tieman reaction	(ii)	$CHCl_3$	(Q)	Reaction involves chain degration
(III)	Aldol condensation	(iii)	KOH	(R)	Synthesis of amine takes place
(IV)	Isocyanide test	(iv)	Br_2	(S)	Used to test primary amines

4. Which of the following option is correctly matched ?

(A) (I), (iv), (P) (B) (II), (iv), (Q) (C) (III), (i), (R) (D) (IV), (ii), (S)

5. Select incorrect match :

(A) (I), (iv), (Q) (B) (II), (i), (R) (C) (III), (iii), (P) (D) (IV), (iii), (S)

6. How many reaction/test from column-I can use KOH as reagent ?

(A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
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Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

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1. Lime stone(X) $\xrightarrow{\Delta}$ Solid (A) + gas (B)

Other Binary Compound of Calcium(Y) $\xrightarrow{H_2O}$ Solution (C) + Gas (D)

Gas (B) turns solution (C) milky but does not react with $KMnO_4/H^+$.

Gas (D) produce white ppt. in tollen reagent and it also produce benzene when passed through red hot iron tube.

The difference of oxidation state of same central atom in (B) & (D) is :

2. How many comparison(s) are incorrect among given ?

(i) H > Li (ionisation energy) (ii) Li > Be (size)

(iii) Na > Rb (size) (iv) O > N (size)

(v) S > P (ionisation energy) (vi) C > O (size)

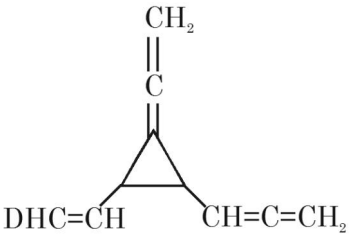
(vii) B > Al (size) (viii) F > Cl (ionisation energy)

(ix) N > O (ionisation energy)

Space for Rough Work

3. Number of non-radioactive alkali metals forming superoxide as major product on heating with excess O_2 = x,
Number of II-A metals of periodic table whose hydrated halides suffer hydrolysis on heating = y.
Number of non radio active alkali metals which dissolve in liquid ammonia and produce blue colour solution = u
The value of (x + y – u) would be....

4. Total number of stereo isomers corresponding to structure



Space for Rough Work

5. The formation of dextrorotatory isomer A from its levorotatory isomer B in water is reversible reaction having 1st order kinetics in both direction. Racemic mixture is the equilibrium product if reaction starts from either A or B. Analysis of polarimeter experiment showed that 10% of B was converted into A in 230.3 sec at 25°C.

(log 5 = 0.7, log 3 = 0.48 and log 2 = 0.3)

Based on given information select number of correct statement(s) among following.

(A) Rate constant of forward reaction at 25°C is $5 \times 10^{-4}\ s^{-1}$

(B) Rate constant of backward reaction at 25°C is $5 \times 10^{-4}\ s^{-1}$

(C) Activation energy of forward reaction is equal to activation energy of backward reaction.

(D) Equilibrium constant (K) at 25°C of B $\rightleftharpoons$ A is 1.

(E) Equilibrium constant increases by increasing temperature.

(F) Molar entropy of A and B at 25°C must be same.

(G) $\Delta_r H$ of the process is zero.

(H) $\Delta_r G^\circ$ of the process is zero.

(I) $\Delta_r G$ of the process $\underset{(l-form)}{B} \underset{k_b}{\overset{k_f}{\rightleftharpoons}} \underset{(d-form)}{A}$ is zero.

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. Let $\vec{a}, \vec{b}, \vec{c}$ be vectors such that $\vec{b} \cdot \vec{c} = 2\vec{a} \cdot \vec{a} = 3\vec{b} \cdot \vec{b} = \vec{a} \cdot \vec{b} \times \vec{c} = 1$ and $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$.
Now if $\vec{a} \times \vec{b}$ is expressed as $x\vec{a} + y\vec{b} + z\vec{c}$, then

(A) $x < 0$ (B) $y < 0$ (C) $z < 0$ (D) $x + y + z < 0$

2. Let f be a twice-differentiable real valued function satisfying the equation :
$x + f(x)\, f'(x) + f'(x) f''(x) = 0$, then

(A) $\lim\limits_{x \to 0} xf(x) = -1$ (B) $\lim\limits_{x \to 0} \dfrac{f'(x)}{\ln|x|} = 0$

(C) $\lim\limits_{x \to 0} xf(x)f'(x) = 1$ (D) $\lim\limits_{x \to 0} \dfrac{f(x)f'(x)}{\ln|x|} = 0$

3. Let the conics $x^2 = xy + 1$ and $xy = x^2 + 1$ have eccentricities e_1 and e_2 respectively then

(A) $e_1 > e_2$ (B) $e_1 < e_2$

(C) $e_1 \cdot e_2 = 2$ (D) $e_1^{-2} + e_2^{-2} = 1$

4. Let $A = [a_{ij}]_{3\times3}$ be a matrix where $a_{ij} \in \{0, 1\}\ \forall\ 1 \le i, j \le 3$. Let $s_i = \sum\limits_{k=1}^{3} a_{ik}$ and $t_j = \sum\limits_{k=1}^{3} a_{kj}$ then the probability that the product $(s_1s_2s_3t_1t_2t_3)$ is an odd integer, is

(A) $\dfrac{1}{2}$ (B) $\dfrac{1}{8}$ (C) $\dfrac{1}{16}$ (D) $\dfrac{1}{32}$

5. If A is an orthogonal matrix of order 3×3 and det.(A) = 1 then which of the following matrices is/are **NOT** invertible

(A) $A - I$ (B) $A^2 - I$

(C) $(A - A^T)^2$ (D) $A - (\det.(A))I$

Space for Rough Work

6. Let for $n \in N$; $(2x^2 - 2x + 1)^n = \sum_{r=0}^{2n} (-1)^r a_r x^r$ then

(A) $\sum_{r=0}^{2n} a_r = 1$

(B) $\sum_{r=1}^{2n} (-1)^r a_r = 0$

(C) $\sum_{k\geq 0} {}^nC_k \, {}^{2n-2k}C_r = 2^{n-r} a_r$

(D) $\sum_{r=0}^{2n} ((-1)^r a_r a_{2n-r}) = 0$ if n = odd

7. Let $(x-k)^2 + y^2 = r^2$, (k, r > 0) is the largest circle, no part of which is lying outside the ellipse $3x^2 + 4y^2 = 12$. Now

(A) if $k < \frac{1}{2}$ then $r + k = 2$

(B) if $k > \frac{1}{2}$ then $r + k = 2$

(C) if $k < \frac{1}{2}$ then $r + k < 2$

(D) if $k > \frac{1}{2}$ then $r + k < 2$

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

Paragraph for Questions 8 - 10

Consider the following path traced by point (x,y) on the curve $x^a + y^a = 1$ lying on the xy plane.

Column-I (Condition)		Column-II (Nature of path)		Column-III (A tangent to the path)	
(I)	a = –1	(i)	Circular	(P)	$x + y = \frac{1}{2}$
(II)	$a = \frac{1}{2}$	(ii)	Hyperbolic	(Q)	x + y = 1
(III)	a = 1	(iii)	Linear	(R)	$x + y = \sqrt{2}$
(IV)	a = 2	(iv)	Parabolic	(S)	x + y = 4

1. Which of the following options is the only **CORRECT** combination?

(A) (III) (iii) (P) (B) (IV) (i) (S) (C) (II) (iv) (P) (D) (I) (ii) (R)

2. Which of the following options is the only **INCORRECT** combination ?

(A) (III) (iii) (Q) (B) (I) (ii) (S) (C) (III) (iii) (P) (D) (II) (iv) (P)

Space for Rough Work

3. Which of the following options is the only **CORRECT** combination?

(A) (II) (iii) (P) (B) (I) (ii) (R) (C) (IV) (i) (Q) (D) (IV) (i) (R)

Paragraph for Questions 11 - 13

Match the various functions **f(x)** given in **Column-I** with their **RANGE** given in **Column-II** and a property of the real number α such that **$f(\alpha) = 1$**, given in **Column-III**.

({· } and [·] respectively denote the fraction part and the greatest integer function)

Column-I (Function)		Column-II (Range)		Column-III ($f(\alpha) = 1$)	
(I)	$f(x) = [x]\{x\}$	(i)	$(-\infty, \infty)$	(P)	$\alpha_{max.} = 1$
(II)	$f(x) = \lvert[x]\rvert\{x\}$	(ii)	$[0, 2)$	(Q)	$\lvert\alpha\rvert_{min.} = \frac{3}{2}$
(III)	$f(x) = \left[x\left[\frac{1}{x}\right]\right]$	(iii)	$[0, \infty)$	(R)	$\lvert\alpha\rvert_{min.} = \frac{5}{2}$
(IV)	$f(x) = \{x\} + \left\{\frac{1}{x}\right\}$	(iv)	$\{k : k \in I, k \geq 0\}$	(S)	$\lvert\alpha - 1\rvert_{min.} = \frac{\sqrt{5}-1}{2}$

4. Which of the following options is the only **CORRECT** combination ?

(A) (I) (iii) (Q) (B) (I) (i) (R) (C) (III) (iv) (R) (D) (II) (iii) (S)

5. Which of the following options is the only **CORRECT** combination ?

(A) (I) (i) (P) (B) (IV) (iii) (S) (C) (III) (ii) (Q) (D) (II) (iii) (Q)

6. Which of the following options is the only **INCORRECT** combination ?

(A) (IV) (ii) (S) (B) (III) (iv) (P) (C) (III) (iv) (S) (D) (I) (i) (R)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

---

1. Let $\tan^{-1}x = \frac{1}{2}\tan^{-1}y = \frac{1}{3}\tan^{-1}z = \frac{\pi}{7}$, then the value of $\frac{1}{2}(x^2 + y^2 + z^2 - x^2y^2z^2)$, is

2. Let for the function $f(x) = 5a^2x^3 + 6ax^2 - 27x + 3b$, $(a \in R^-, b \in R)$, $f'\left(\frac{-5}{9}\right) = 0$, $f''\left(\frac{-5}{9}\right) < 0$ and there exists some $\alpha \neq \frac{-5}{9}$ such that $f'(\alpha) = 0$, $f(\alpha) > 0$. The minimum possible integral value of $(a + b)$, is

3. Let $I = \int_{-3}^{-2} \left(\frac{x^2 - x}{x^3 - 3x + 1}\right)^2 \left(1 + \frac{1}{x^2} + \frac{1}{(1-x)^2}\right) dx$ then the greatest integer less than the value of I, is

4. Let f be a function such that $xf(x) = \pi\sin x \ \forall\ x \in (0, \pi)$ and $f(x) + f(\pi - x) = g(x)$ then $g_{max.}$ is

5. Let $f(x) = (x - 1)\,|\text{sgn}(x)|$ and $g(x) = \max.(f(t))$, $-\infty < t \leq x$, $-\infty < x < \infty$, then the number of points where g(x) is non-differentiable is (sgn (x) represents signum function of x)

Space for Rough Work

Space for Rough Work

PHYSICS

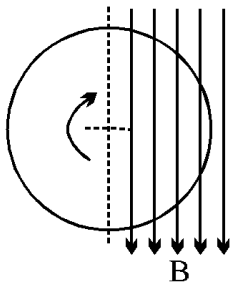
SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

Space for Rough Work

1. A thin non conducting disc of radius R is rotating clockwise (see figure) with an angular velocity ω about its central axis, which is perpendicular to its plane. Both its surfaces carry +ve charges of uniform surface density. Half the disc is in a region of a uniform, unidirectional magnetic field B parallel to the plane of the disc, as shown. Then,



(A) The net torque on the disc is zero.

(B) The net torque vector on the disc is directed leftwards.

(C) The net torque vector on the disc is directed rightwards.

(D) The net torque vector on the disc is parallel to B.

2. At the initial moment three point A, B and C are on a horizontal plane along a straight line such that AB = BC. Point A begins to move upward with a constant velocity 'v' and point C downward without any initial velocity at a constant acceleration 'a'. If the points begin to move simultaneously, then the initial velocity and acceleration of point B for all the three particles to be constantly on same straight line must be:

(A) $\frac{v}{2}$upwards, $\frac{a}{2}$downwards

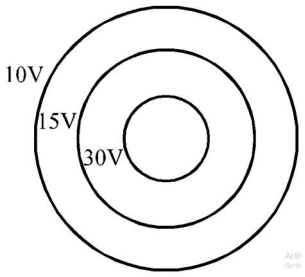
(B) $\frac{v}{2}$upwards, $\frac{a}{2}$upwards

(C) $\frac{v}{2}$downwards, $\frac{a}{2}$downwards

(D) $\frac{v}{2}$downwards, $\frac{a}{2}$upwards

Space for Rough Work

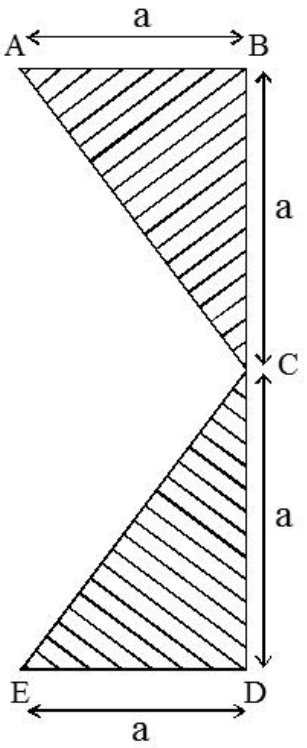
3. The diagram shows three concentric conducting spherical shells having radii R, 2R and 3R. The initial potential of each shell is as mentioned in the figure. If the inner most shell is earth then the charge present on its outer surface would be equal to



(A) zero (B) $40\pi\varepsilon_0 R$ (C) $4\pi\varepsilon_0 R$ (D) $-\pi\varepsilon_0 R$

Space for Rough Work

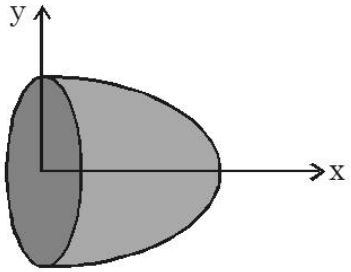
4. ABCDECA is a planar body of mass m of uniform thickness and same material. The dimensions are as shown in the figure. The moment of inertia of the body about an axis passing through point A and perpendicular to planar body is I_1 and that of about an axis passing through C and perpendicular to planer body is I_2. Then choose the correct relation :



(A) $\frac{I_1}{I_2} = 3$ (B) $\frac{I_1}{I_2} = \frac{1}{3}$ (C) $\frac{I_1}{I_2} = 1$ (D) $\frac{I_2}{I_1} = \frac{2}{3}$

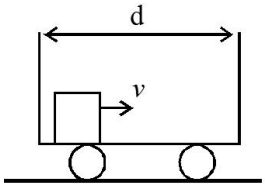
Space for Rough Work

5. If an ellipse is rotated about X-axis an ellipsoid is formed, then the coordinates of the centre of mass of a solid uniform semi–ellipsoid formed by rotating a quarter ellipse given by the equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ as shown in diagram is :-



(A) $\left(\frac{3a}{8},0\right)$ (B) $\left(\frac{a}{8},0\right)$ (C) $\left(\frac{a}{4},0\right)$ (D) $\left(\frac{2a}{3},0\right)$

6. In a smooth stationary cart of length d, a small block is projected from the rear end of the cart along it's length with velocity v towards front end. Coefficient of restitution for each collision is e. The cart rests on a smooth ground and can move freely. The time taken by block to come to rest w.r.t. cart is (Assume the time for collision is negligibly small).



(A) $\frac{ed}{(1-e)v}$ (B) $\frac{ed}{(1+e)v}$ (C) $\frac{d}{e}$ (D) Infinite

Space for Rough Work

7. According to Moseley's law the ratio of the slopes of graph between $\sqrt{f}$ and Z for K_β and K_α is: (where the symbols have their usual meaning).

(A) $\sqrt{\frac{32}{27}}$ (B) $\sqrt{\frac{27}{32}}$ (C) $\sqrt{\frac{33}{22}}$ (D) $\sqrt{\frac{22}{33}}$

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. A micrometer has a resistance of 100Ω and a full scale range of 50μA. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).

(A) 50 V range with 10kΩ resistance in series

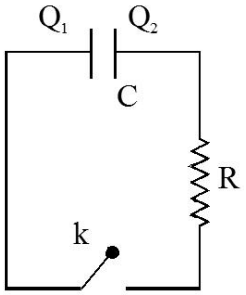
(B) 10 V range with 200kΩ resistance in series

(C) 5mA range with 1Ω resistance in parallel

(D) 10mA range with 1kΩ resistance in parallel

Space for Rough Work

2. The two plates of an ideal capacitor of capacitance C are given charges Q_1 and Q_2 as shown in the circuit below. This capacitor is connected across a resistance R as shown key is closed at t = 0. Then after time t.



(A) Total charge on the right plate is $q_2 = \frac{Q_1 + Q_2}{2} - \left(\frac{Q_1 - Q_2}{2}\right) e^{-\frac{t}{RC}}$

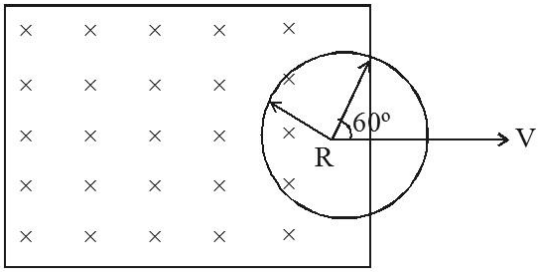
(B) Total charge on the left plate is $q = \frac{Q_1 + Q_2}{2} + \left(\frac{Q_1 - Q_2}{2}\right) e^{-\frac{t}{RC}}$

(C) Initial potential difference across the plates is given by $\left|\frac{Q_1 - Q_2}{2C}\right|$

(D) Initial potential difference across the plates is given by $\frac{Q_1 + Q_2}{2C}$

Space for Rough Work

3. A circular conducting loop of radius R and resistance per unit length λ is pulled out from the region of uniform magnetic field with constant velocity v. The situation shown in the figure corresponds to time t = 0. Mark out the correct statement(s)



(A) Just after t = 0 i.e., the motion starts, the induced current in the loop is $\frac{\sqrt{3}Bv}{2\pi\lambda}$

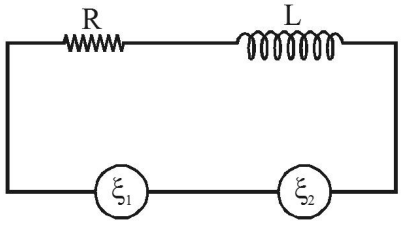
(B) Current will be induced in the loop for $\frac{3R}{2v}$ sec

(C) At any time t, the current induced in the loop is given by $\frac{Bv\sqrt{3R^2 - 4v^2t^2 + 4Rvt}}{\lambda \times 2\pi R}$ and is in clockwise direction.

(D) Induced current is in clockwise direction for t = 0 to $\frac{R}{2v}$ and thereafter it becomes in anticlockwise direction.

Space for Rough Work

4. Two AC sources $\xi_1 = 10 \sin(\omega t)V$ and $\xi_2 = 10\sin\left(\omega t - \frac{\pi}{3}\right)V$ are fed to an LR circuit. The resistance is $R = 10\ \Omega$ and inductive reactance is $X_L = 10\sqrt{3}\Omega$:-



(A) The equivalent emf of the AC source has a peak value of $10\sqrt{3}V$.

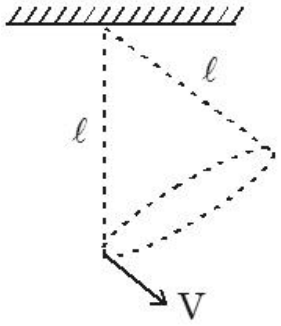
(B) The peak value of current is $\frac{\sqrt{3}}{2}A$

(C) The voltage across the inductor at $t = \frac{\pi}{3\omega}$ is $\frac{15\sqrt{3}}{2}V$

(D) The voltage across the resistor at $t = \frac{\pi}{3\omega}$ is $\frac{15}{2}V$

Space for Rough Work

5. A positively charged particle of charge q and mass m is suspended from a point by a string of length ℓ in a vertical plane. In the entire space a uniform horizontal electric field E exists. The particle is drawn aside so that string becomes vertical and then it is projected horizontally with velocity v such that particle starts to move along a circle with some constant speed v. Then choose the correct statements.



(A) Speed of the particle is $v = \frac{qE}{m}\sqrt{\frac{\ell}{g}}$

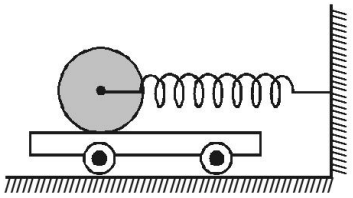
(B) Tension in the string is $mg + \frac{q^2E^2}{mg}$

(C) Time period of revolution is $2\pi\sqrt{\frac{\ell g}{g^2 + \left(\frac{qE}{m}\right)^2}}$

(D) Time period of revolution is independent of mass of particle.

Space for Rough Work

6. In the figure shown, the mass of the disc as well as that of the trolley is M. The spring is ideal and has stiffness k. The trolley can move horizontally on the floor without friction, and the disc can roll on the trolley surface without slipping. The spring is compressed and the system released so that oscillations begin. At the instant of release, magnitude of :



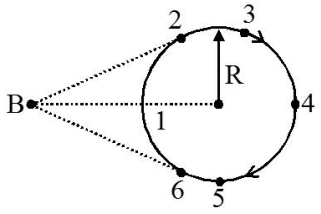
(A) Acceleration of center of disc = twice of that of trolley

(B) Acceleration of center of disc = thrice of that of trolley

(C) Spring force is four times of friction.

(D) Spring force is two times of friction.

7. An isotropic sound source A is moving in a circle of radius R with small speed v. An observer B is hearing this sound (see figure).



(A) The intensity of sound heard by B will be maximum just after A crosses 1.

(B) The frequency of sound heard by B will be minimum when A is at 6.

(C) The frequency of sound heard by B will be minimum just after A crosses 2.

(D) The frequency heard by B when source is at 1 & 4 is minimum.

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

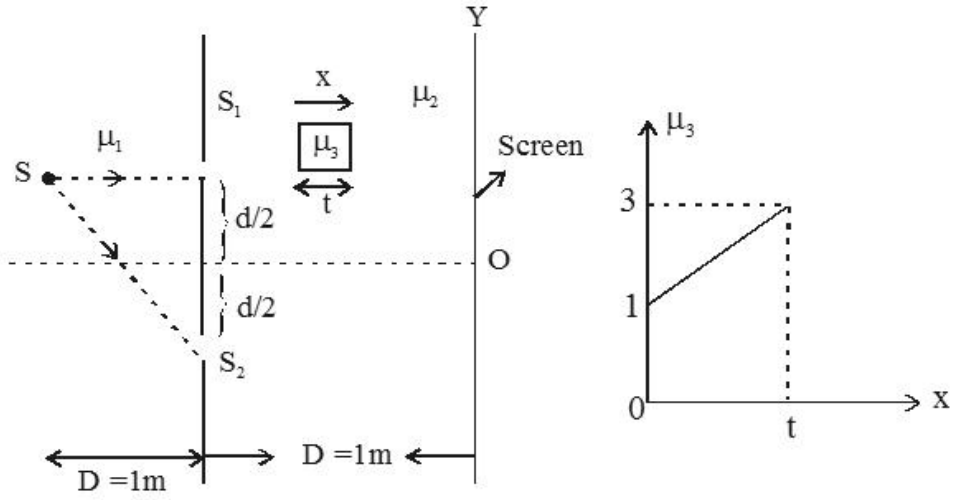
Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 In all other cases.

Space for Rough Work

Paragraph for Questions 15 and 16

In YDSE arrangement shown in figure, fringes are seen on screen using monochromatic source S having wavelength 3000Å (in air). S_1 and S_2 are two slits separated by d = 1mm from each other and D = 1m. Left of slits S_1 and S_2 medium of refractive index $\mu_1 = 2$ is present & to the right of slits S_1 and S_2 medium of refractive index $\mu_2 = \frac{3}{2}$ is present. A thin slab of thickness 't' is placed in front of S_1. The refractive index of slab μ_3 varies with distance (x) from its starting face as shown in figure.



1. In order to get central maxima at the centre of the screen (O), the thickness of slab required is :-

(A) 1μm (B) 2μm (C) 0.5μm (D) 1.5 μm

2. If the thickness of the slab is selected 1mm, then the position of the central maxima will be (Y - coordinate) Taking O to be origin

(A) $\frac{1}{3}$mm (B) $-\frac{1}{3}$mm (C) $\frac{1}{6}$mm (D) $-\frac{1}{6}$mm

Space for Rough Work

Paragraph for Questions 17 and 18

Air Standard Otto cycle :

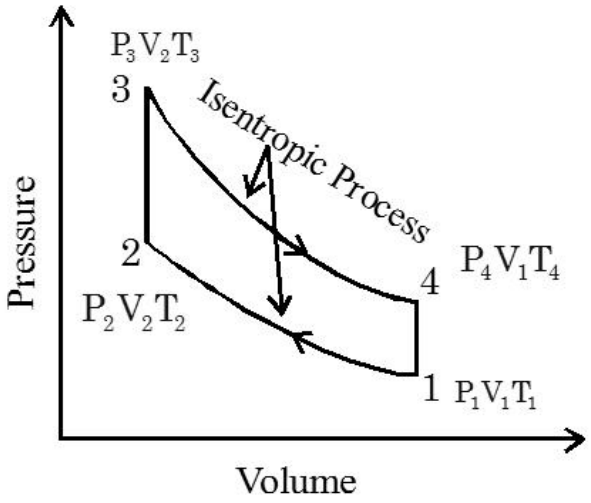
The air-standard-Otto cycle is the ideallzed cycle for the spark-ignition internal combustion engines. This cycle is shown on P-V and T-S diagrams. The Otto cycle 1-2-3-4 consists of following four process:

Process 1-2: Reversible adiabatic compression of air.

Process 2-3: Heat addition at constant volume.

Process 3-4: Reversible adiabatic expansion of air.

Process 4-1: Heat rejection at constant volume.



Mean Effective Pressure is defined as the ratio of the net work done to the displacement volume of the piston.

3. For a given mass of working gas of adiabatic constant $\gamma \left(= \frac{C_P}{C_V}\right)$ if the compression ratio r is defined as $r = \frac{V_1}{V_2}$ (Where V_1 is volume of gas at 1 and V_2 is volume of gas at 2). Then the efficiency of the cycle is given as.

(A) $n_{th} = 1 - \left(\frac{1}{r}\right)^{\gamma-1}$

(B) $n_{th} = 1 + \left(\frac{1}{r}\right)^{\gamma-1}$

(C) $n_{th} = 1 + \left(\frac{1}{r}\right)^{\gamma+1}$

(D) $n_{th} = 1 - \left(\frac{1}{r}\right)^{\gamma+1}$

Space for Rough Work

4. If the displacement volume is given by $(V_1 - V_2)$, then the mean effective pressure of the above cycle is given as

(A) $\left(\frac{1}{\gamma-1}\right)\left(\frac{P_1}{T_1}\right)\left(\frac{r}{r-1}\right)\{(T_3-T_2)-(T_4-T_1)\}$

(B) $\left(\frac{1}{\gamma+1}\right)\left(\frac{P_1}{T_1}\right)\left(\frac{r}{r-1}\right)\{(T_3-T_2)+(T_4-T_1)\}$

(C) $\left(\frac{1}{\gamma-1}\right)\left(\frac{P_1}{T_1}\right)\left(\frac{r}{r+1}\right)\{(T_3-T_2)+(T_4-T_1)\}$

(D) $\left(\frac{1}{\gamma-1}\right)\left(\frac{P_1}{T_1}\right)\left(\frac{r}{r-1}\right)\{(T_3+T_2)+(T_4-T_1)\}$

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. Choose correct Statement:

(A) In group 15 Hydrides, NH_3 is least volatile.

(B) HNO_2 can act as oxidiser as well as reducer.

(C) NH_3 is a non-polar molecule.

(D) N_2 is a reactive gas at room temperature.

2. Choose incorrect statement :

(A) SnF_4 is more ionic than SnI_4.

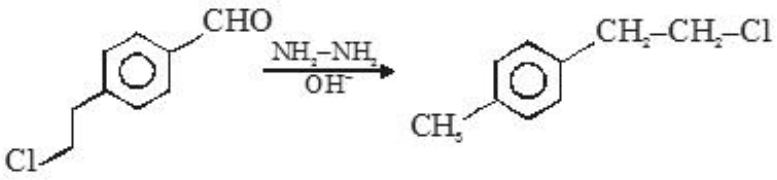
(B) Cement contains silicates.

(C) Diamond does not conduct electricity.

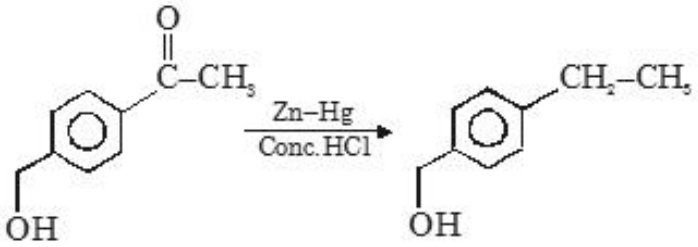
(D) CCl_4 easily hydrolyses to give CH_4 at room temperature.

Space for Rough Work

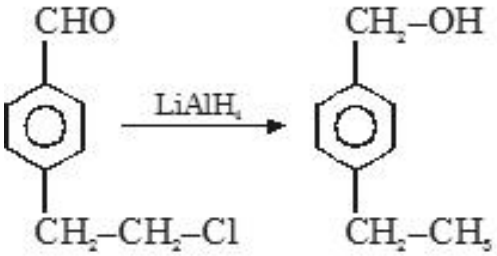
3. In given reactions which one is correct ?

(A) 

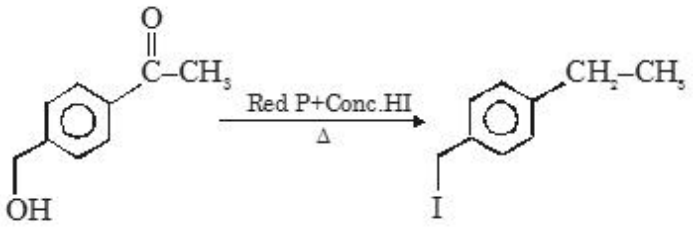
(major)

(B) 

(major)

(C) 

(major)

(D) 

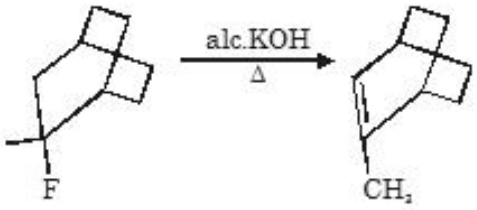
(major)

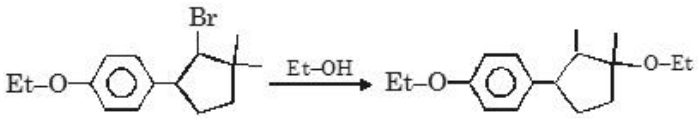
Space for Rough Work

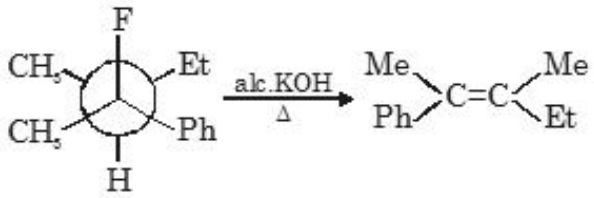
4. Which compound will not give positive $NaHCO_3$ test ?

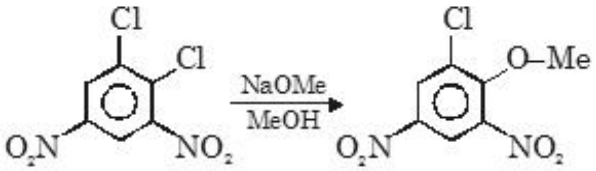
(A) Picric acid
(B) Benzoic acid
(C) Carbinol
(D) Terephthalic acid

5. Identify reaction that correctly match with their major product.

(A) 

(B) 

(C) 

(D) 

Space for Rough Work

6. Which of the following statements is correct ?

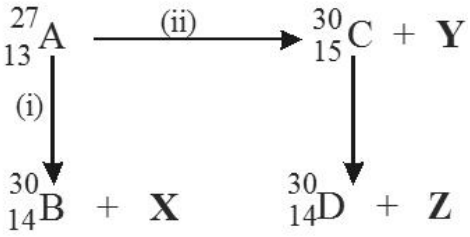
(A) Milk is not an emulsion.

(B) Lyophilic hydrosols have lower surface tension than H_2O.

(C) Lyophobhic hydrosols have lower surface tension than water.

(D) Conductivity of Lyophilic colloids is negligible

7. Bombardment on nucleus A by α-particle leads to its artificial disintegration in two ways, (i) and (ii) as shown. Products X, Y and Z respectively are :



(A) proton, neutron, positron

(B) neutron, positron, proton

(C) proton, positron, neutron

(D) positron, proton, neutron

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

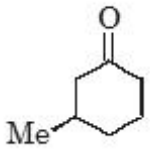
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

Space for Rough Work

1. Which carbonyl compound(s) give racemic cyanohydrin, when react with HCN.

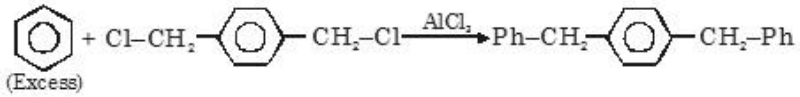
(A) Acetone

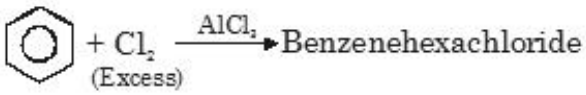
(B) Cyclopentanone

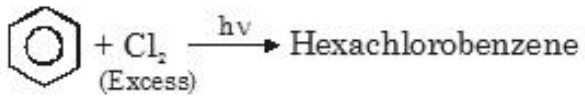
(C) Equal amount of (±) 

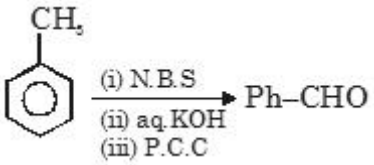
(D) Acetaldehyde

2. In given reaction, which one is/are correct according to major product.

(A) 

(B) 

(C) 

(D) 

Space for Rough Work

3. Choose correct statement(s)

(A) Hydrolysis of Mg_2C_3 involves redox reaction.

(B) NF_3 does not undergo hydrolysis in ordinary conditions.

(C) $B_3N_3H_6$ produces no gaseous product when hydrolysed at high temperature

(D) The products of hydrolysis of hypophosphoric acid ($H_4P_2O_6$) are H_3PO_3 and H_3PO_4.

4. Which of the following reaction(s) will produce at least one gaseous compound ?

(A) $Na_2CO_3(aq) + dil.H_2SO_4 \longrightarrow$

(B) $Na_2SO_3(aq) + dil.H_2SO_4 \longrightarrow$

(C) $Na_2C_2O_4(aq) + conc.H_2SO_4 \longrightarrow$

(D) $NaNO_3(aq) + conc.H_2SO_4 \xrightarrow{\Delta}$

5. Anhydrous $CaCl_2$ is often used as desiccant. In the presence of excess $CaCl_2$ the amount of water taken up is governed by $K_P = 6.4 \times 10^{85}$ for the following reaction at room temperature.

$$CaCl_2(s) + 6H_2O(g) \rightleftharpoons CaCl_2 \cdot 6H_2O(s)$$

Which of the following statements is correct about above reaction ?

(A) At equilibrium addition of $CaCl_2$ will shift reaction in forward direction.

(B) Vapour pressure of water at equilibrium in a closed vessel that contains $CaCl_2(s)$ is 5×10^{-15} atm approximately.

(C) Vapour pressure of water at equilibrium in a closed vessel becomes half if volume of container is reduced to half at same temperature.

(D) If temperature is increased, then vapour pressure of water at equilibrium in closed vessel decreases.

Space for Rough Work

6. Which of the following statements is/are correct ?

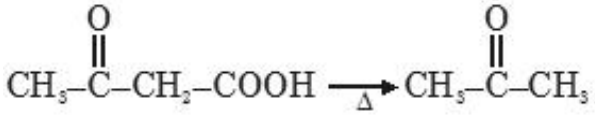
(A) At constant volume, there will be no effect of changing T or P on mean free path.

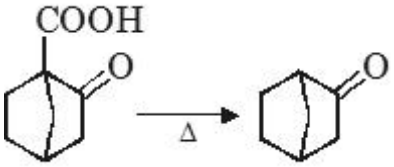
(B) Collision number increases on increasing temperature at constant volume.

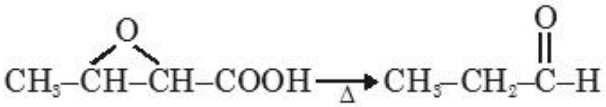
(C) If diffusion coefficient is proportional to mean free path and mean speed then on increasing absolute temperature of an ideal gas by four times and pressure by two times, diffusion coefficient of gas increases four times.

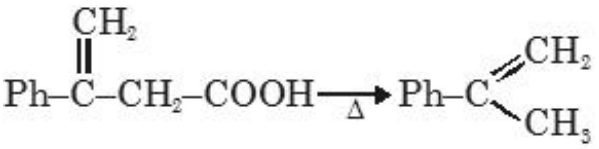
(D) According to kinetic theory of gases, between two successive colloisions, a gas molecule travels in straight line path.

7. Which reaction is/are correct ?

(A) 

(B)  (major)

(C) 

(D) 

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

d-Block metals are extracted from their ores by a wide variety of techniques. Most metals are obtained by direct treatment of their ores with chemical reagents, but the extraction of certain metals require electrolysis . Pyrolusite (MnO_2) Hausmannite (Mn_3O_4) Rhodocroside ($MnCO_3$) are the natural abundent ore of Mn use to extract metal and pyrolusite (MnO_2) ore is used to prepare some compounds, such as $KMnO_4$ can be prepared with intermediate fomation of K_2MnO_4.

1. The set of metals whose extraction involves hydrometallurgy is:

(A) Al, Zn (B) Fe, Cu (C) Ag, Au (D) Fe, Ag

Space for Rough Work

2. Consider following Statements

I: $KMnO_4$ can act as oxidiser in acidic, basic and neutral medium.

II: MnO_4^{2-} is paramagnetic while MnO_4^{-1} is diamagnetic.

III. MnO_4^- is only oxidising agent but MnO_4^{2-} can act as oxidising as well as reducing agent

IV. Mn_3O_4 can be converted to Mn by aluminothermite process

The Correct statement is/are:

(A) I, II & III only (B) I, II & IV only

(C) I, II, III & IV (D) I & IV only

Paragraph for Questions 17 and 18

The electronic ground state of hydrogen atom contains one electron in first orbit. If sufficient energy is provided, the electron can be promoted to higher energy level. The electronic energy of hydrogen like species (any atom/ions with nuclear charge |Ze| and one electron) can be given as

$$E_n = -R_H \frac{Z^2}{n^2}$$

Where

R_H = Rydberg's constant

n = Principle quantum number , Z = Atomic number

3. The ratio of energy of an electron in the ground state of Be^{3+} ion to that of ground state of H-atom is

(A) 16 (B) 4 (C) 1 (D) 8

4. The experimental $I.E_1$ of He atom is 24.58 eV. Total energy required to remove two electrons from He is

(A) 49.16 eV (B) 78.98 eV (C) 29.82 eV (D) None of these

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 21)

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Negative Marks : –1 In all other cases

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1. If line $x = y = z$ intersect the line $x\sin A + y\sin B + z\sin C - 2d^2 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - d^2$ then $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ equals, where $(A + B + C) = \pi$

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{32}$ (D) $\frac{1}{12}$

2. Solution of $\frac{dy}{dx} = \frac{y^3}{e^{2x} + y^2}$ is

(A) $e^{-2x}y^2 + 2\ln|y| = C$ (B) $e^{2x}y^2 - 2\ln|y| = C$

(C) $e^x + \ln|y| = C$ (D) None of these

3. Let $f(x) = x^2 - 2x$, $g(x) = f(f(x) - 1) + f(5 - f(x))$ then

(A) $g(x) < 0\ \forall\ x \in R$ (B) $g(x) < 0$ for some $x \in R$

(C) nothing can be said about $g(x)$ (D) $g(x) \geq 0\ \forall\ x \in R$

4. If $[\bar{a}\ \bar{b}\ \bar{c}] = |\bar{d}| = 6$, then $\left|[\bar{d}\ \bar{b}\ \bar{c}]\,\bar{a} + [\bar{d}\ \bar{c}\ \bar{a}]\,\bar{b} + [\bar{d}\ \bar{a}\ \bar{b}]\,\bar{c}\right| =$

(A) 36 (B) 12 (C) 24 (D) 6

Space for Rough Work

5. Line segments drawn from vertex opposite to the hypotenuse of a right angle triangle to the points trisecting the hypotenuse, have lengths sinx and cosx $\left(0 < x < \frac{\pi}{2}\right)$. Then length of hypotenuse is

(A) $\frac{1}{3}$ (B) $\frac{3}{2}$

(C) $\frac{3\sqrt{5}}{5}$ (D) can't be determined

6. Let $x + y = 0$ and $x - y = 0$ are tangents drawn to a parabola whose focus is at S(1, 2) then length of latus rectum of parabola is

(A) $\frac{6}{\sqrt{5}}$ (B) $\frac{4}{\sqrt{5}}$ (C) $\frac{2}{\sqrt{5}}$ (D) $\frac{1}{\sqrt{5}}$

7. If A is an involutary matrix and diagonal matrix of non positive entries and order 3 then

(A) there may exist some diagonal element in A, which is/are zero

(B) $|A| = -1$

(C) A^{-1} does not exist

(D) $|3\text{adj}.2A| = -1728$

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. If $4a^2 - 5b^2 + 6a + 1 = 0$ and the line $ax + by + 1 = 0$ touches a fixed circle, then

(A) centre of circle is (3, 0)

(B) radius of circle = $\sqrt{5}$

(C) circle passes through (0, 1)

(D) None of these

Space for Rough Work

2. Number of real solution(s) of $2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} = 3$ is less than

(A) 0 (B) 1 (C) 2 (D) 3

3. Let $\int_0^x \frac{(mt\cos 4t - \ell \sin 4t)dt}{t^2} = \frac{\ell \sin 4x}{x} - 1$, where $\left(0 < x < \frac{\pi}{4}\right)$, then

(A) $\ell + m = \frac{5}{4}$ (B) $m = 4\ell$ (C) $\ell - m = \frac{3}{4}$ (D) $m = \ell + \frac{1}{4}$

4. If tangent of any member of family of hyperbola $xy = 4\sin^2\theta$, $\theta \in (0, 2\pi) - \{\pi\}$ is not a normal to member of family of circles $x^2 + y^2 - 2x - 2y + \mu = 0$, μ is real parameter, then θ belongs to

(A) $\left(\frac{5\pi}{6}, \frac{7\pi}{6}\right)$ (B) $\left(0, \frac{\pi}{6}\right)$ (C) $\left(\frac{11\pi}{6}, 2\pi\right)$ (D) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$

5. $f(x) = [\tan^{100}x]$ where $[\cdot]$ greatest integer function. Then

(A) f(x) is discontinuous at $x = \tan^{-1}(3^{\frac{1}{100}}), \tan^{-1}(100^{\frac{1}{100}})$

(B) f(x) discontinuous at $x = \tan^{-1}(3^{\frac{1}{100}}), \tan^{-1}(99^{\frac{1}{100}})$

(C) $\int_0^{\tan^{-1}\left(4^{\frac{1}{100}}\right)} f(x)dx = 3\tan^{-1}\left(4^{\frac{1}{100}}\right) - \tan^{-1}\left(3^{\frac{1}{100}}\right) - \tan^{-1}\left(2^{\frac{1}{100}}\right) - \frac{\pi}{4}$

(D) $\int_0^{\tan^{-1}\left(4^{\frac{1}{100}}\right)} f(x)dx = 3\tan^{-1}4^{\frac{1}{100}} - 2\tan^{-1}3^{\frac{1}{100}}$

Space for Rough Work

6. Area of region given by curves $1 - y^2 \geqslant |x|$ and $|x| + |y| \geqslant 1$ is A and area of region $\{(x, y) : x^2 \leqslant y \leqslant |x|\}$ is B. Then $\frac{A}{B}$ is less than

(A) 1 (B) 2 (C) 3 (D) 4

7. Set of all real values of p such that both the roots of the equation $(p-5)x^2 - 2px + (p-4) = 0$ are positive, one is less than 2 and other is lying between 2 and 3 is

(A) $\left(\frac{49}{4}, 24\right)$ (B) $(5, \infty)$

(C) $(-\infty, 4) \cup \left(\frac{49}{4}, \infty\right)$ (D) $\left(4, \frac{49}{4}\right)$

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

Let A = {1, 2, 3, ... n} is a set containing n elements.

1. For any given n, number of subsets of A having k(k < n) as largest element, is

(A) 2^k (B) $n-k$

(C) 2^{k-1} (D) None of these

2. The number of subsets of A having at least one odd integer is
[where [.] represents greatest integer function]

(A) $2^{\left[\frac{n}{2}\right]}$ (B) $2^{\left[\frac{n+1}{2}\right]}$ (C) $2^n - 2^{\left[\frac{n}{2}\right]}$ (D) $2^{\left[\frac{n}{2}\right]+1}$

Paragraph for Questions 17 and 18

Let S be the set of points $\{(x, y) : 0 \le x \le 3, 0 \le y \le 4 \text{ where } x, y \in I\}$ two distinct points are randomly chosen from S.

3. The probability that mid point of the line segment of two points chosen belongs to S is

(A) $\frac{21}{95}$ (B) $\frac{42}{95}$ (C) $\frac{36}{95}$ (D) None of these

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Space for Rough Work

4. If two points (x_1, y_1) and (x_2, y_2) are chosen such that $x_1 = x_2$ and $y_1 \neq y_2$ then probability that mid point of these two points belongs to S, is

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{8}{15}$ (D) $\frac{8}{95}$

Space for Rough Work

Space for Rough Work

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
12-06-2021

JEE(Main + Advance) : JEE (MAIN+ADVANCED) NURTURE PHASE-I & II_MT#03 (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 12-06-2021

ANSWER KEY — PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	A,B,D	B,C,D	A,B,D	B,C,D	A,B	A,B
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	A	D	D	A	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	2	9	5	2		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	A,B,C,D	A,B,C,D	A,C,D	A,B,D	A,B,C	A,B,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	D	B	A	D	B	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	4	0	8	7		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	B,D	A,D	D	A,B,C,D	B,C,D	B,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	C	D	B	D	C	
SEC-3	Q.	1	2	3	4	5		
	A.	7	6	5	4	2		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (B,D)

$r \propto \frac{1}{m}$

so m double, radius half

$E \propto m$

so IE will double and gap between levels will also double.

2. Ans (A,B,D)

When water is in liquid state

$$w_g = \frac{3}{2}mgh = k = \left[\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{3}mr^2\right)\frac{v^2}{r^2}\right] + \left[\frac{1}{2}\left(\frac{m}{2}\right)v^2\right]$$

water will have only translational KE

when water is freezed

$$w_g = k' = \frac{3}{2}mgh = \frac{1}{2}mv'^2 + \frac{1}{2}\left(\frac{2}{3}mr^2\right)\frac{v'^2}{r^2} + \left[\frac{1}{2}\frac{m}{2}v'^2 + \frac{1}{2}\frac{2}{5}\frac{m}{2}r^2\frac{v'^2}{r^2}\right]$$

water will have translational as well as rotational KE so $v' < v$

3. Ans (B,C,D)

$\rho = kr^2 = \frac{\rho_0}{R^2} r^2$

(A) $v_{esc} = \sqrt{\frac{2GM}{R}}$

$M = \int_0^R \frac{\rho_0}{R^2} r^2 4\pi r^2 dr = \frac{\rho_0 4\pi}{R^2} \frac{R^5}{5} = \frac{\rho_0 4\pi R^3}{5}$

so $v_{esc} = \sqrt{\frac{2G\rho_0 4\pi R^3}{5R}} = \sqrt{\frac{8\pi G\rho_0 R^2}{5}}$

(B) $g = \frac{GM}{r^2} = \frac{G}{r^2} \int_0^r \frac{\rho_0}{R^2} r^2 4\pi r^2 dr =$

$\frac{G\rho_0 4\pi r^3}{5R^2}$

$g \propto r^3$

(C) as $v_{esc} = \sqrt{\frac{2GM}{R}}$ so will be same for other planet also.

(D) Energy required $= \frac{1}{2} m v_{esc}^2 \propto m$

so $\frac{E_1}{E_2} = \frac{1}{2}$

4. Ans (A,B,D)

$V_A = \frac{Kq_A}{R} + \frac{Kq_B}{2R} = 2V,$

$V_B = \frac{K(q_A + q_B)}{2R} = \frac{3}{2}V$

$\Rightarrow \frac{q_A}{q_B} = \frac{1}{2}$

After earthing of B, $q_A = q_B$ &

$V'_B = \frac{K\left(q'_A + q'_B\right)}{2R} = 0$

$\Rightarrow \left|\frac{q'_A}{q_B}\right| = 1 \ \& \ V'_A - V'_B = \frac{V}{2}$

5. Ans (B,C,D)

$|E| = nat^{n-1}$

6. Ans (A,B)

$P =$

$\frac{P_0}{2}\left[\sin\left(203\pi t + \frac{5\pi}{6}\right) + \sin\left(197\pi t + \frac{\pi}{6}\right)\right]$

$f_1 = \frac{203\pi}{2\pi} > 100$

$f_2 = \frac{197\pi}{2\pi} < 100$

$f_{beat} = \frac{203\pi}{2\pi} - \frac{197\pi}{2\pi} = 3$

7. Ans (A,B)

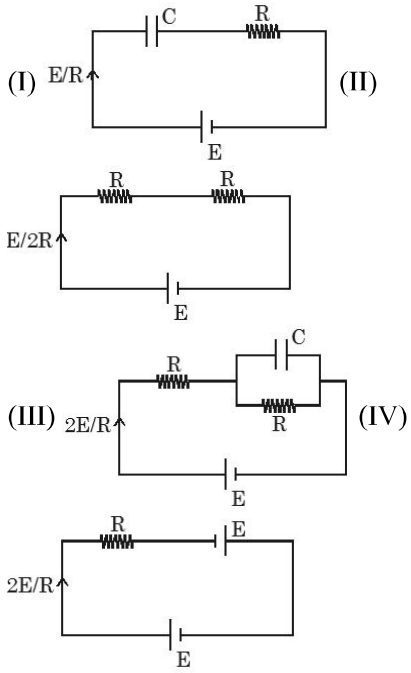
$\delta_{net} = \delta_1 + \delta_2 + ... + \delta_n$

where all have same magnitude and alternate deviations are in opposite direction.

PHYSICS

SEC-2

1. **Ans (B)**



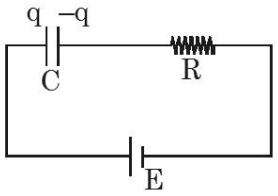
Current in III & IV is maximum

in (IV) no energy is stored as no capacitor

so ans (B) IV, i, Q

2. **Ans (A)**

(I) At t = RC

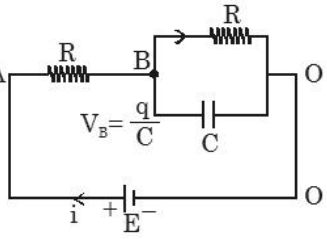


$q = CE(1 - e^{-1})$ = charge drawn from battery

$= 0.63\ CE$

(II) Charge drawn = $iRC = \frac{E}{2R}RC = 0.5CE$

(III)



$q = \frac{CE}{2}\left(1 - e^{\frac{-2t}{RC}}\right)$ $\quad V_B = \frac{q}{C}$ & $V_A = E$

$i = \frac{v_A - v_B}{R} = \frac{E}{R} - \frac{E}{2R}\left(1 - e^{\frac{-2t}{RC}}\right)$

$= \frac{E}{2R} + \frac{E}{2R}e^{\frac{-2t}{RC}}$

Charge drawn = $\int_0^{RC} idt =$

$\frac{E}{2R}\int_0^{RC}\left(1 + e^{\frac{-2t}{RC}}\right)dt$

$= \frac{E}{2R}\left[\left(t + \frac{e^{-\frac{2t}{RC}}}{(-2)}(RC)\right)\right]_0^{RC}$

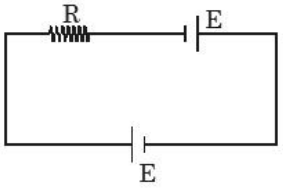
$= \frac{E}{2R}\left[\left(RC - \frac{RC}{2}e^{-2}\right) - \left(0 - \frac{RC}{2}\right)\right]$

$= \frac{E}{2R}\left[\frac{3RC}{2} - \frac{RC}{2}e^{-2}\right] =$

$\frac{E}{2R}\frac{RC}{2}\left[3 - e^{-2}\right]$

$= \frac{CE}{4}\left(3 - e^{-2}\right) = 0.72\ CE$

(IV)



charge drawn = $\frac{2E}{R}RC = 2CE$

3. Ans (D)

in (III) $i_{t=0} = \frac{E}{R}$

$i_{t=\infty} = \frac{E}{2R}$

4. Ans (D)

(I) F_{ABCD} due to $\rho_1 = \left(\rho_1 g \frac{h}{2}\right) h\ell$;

(II) F_{ABCD} due to $\rho_2 = 0$

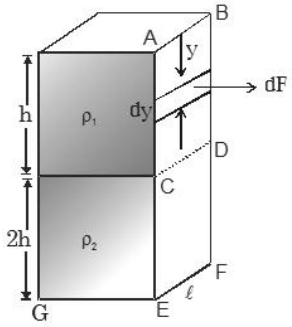
(III) F_{CDEF} due to $\rho_1 = \rho_1 gh(2h\ell)$;

(IV) F_{CDEF} due to $\rho_2 = (\rho_2 gh)(2h\ell)$

Pressure at face CDEF due to ρ_1 is constant to effectively acts at centre

5. Ans (A)

(I)



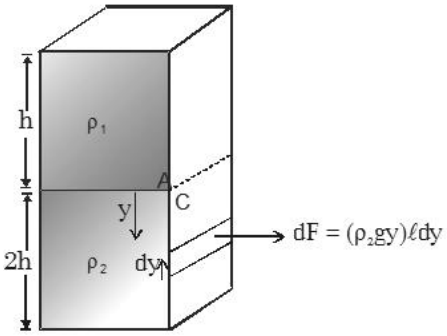
$$\int d\tau = \int_0^h (\rho_1 gy)\,\ell dy\,(3h - y)$$

$$\tau = \rho_1 g\ell \left[\frac{3hy^2}{2} - \frac{y^3}{3}\right]_0^h = \rho_1 g\ell \frac{7}{6}h^3$$

(II) $\tau = 0$

(III) $\tau = (\rho_1 gh)(2h\ell)(h) = 2\rho_1 gh^3\ell$

(IV)



$$\int d\tau = \int_0^{2h} (\rho_2 g\ell y dy)\,(2h - y)$$

$$\Rightarrow \tau = \rho_2 g\ell \left[2h\frac{y^2}{2} - \frac{y^3}{3}\right]_0^{2h}$$

$$= \rho_2 g\ell \left[h.4h^2 - \frac{8h^3}{3}\right] =$$

$$\rho_2 g\ell \left[\frac{4h^3}{3}\right] = \frac{4\rho_2 g\ell h^3}{3}$$

so $\frac{7}{6}\rho_1 g\ell h^3 < \frac{4}{3}\rho_2 g\ell h^3$ (as $\rho_1 = \rho_2$)

so minimum torque in (I)

6. Ans (D)

Weight of liquid $\rho_2 = \rho_2(2h\ell h)g = 2\rho_2 \ell h^2 g$

PHYSICS

SEC-3

1. Ans (5)

From COAM

$$mv_0\frac{\ell}{6} = \left(2m\left(\frac{\ell}{3}\right)^2 + m\left(\frac{2\ell}{3}\right)^2\right)\omega$$

$$\omega = \frac{v_0}{4\ell}$$

From COM

$mv_0 = 3mv_{cm}$

$$v_{cm} = \frac{v_0}{3}$$

Loss in KE

$$= \frac{1}{2}mv_0^2 - \left\{\frac{1}{2}(3m)v_{cm}^2 + \frac{1}{2}I_{cm}\omega^2\right\}$$

$$= \frac{1}{2}mv_0^2 - \left\{\frac{1}{2}(3m)\left(\frac{v_0}{3}\right)^2 + \frac{1}{2}\left(\frac{2m\ell^2}{3}\right)\left(\frac{v_0}{4\ell}\right)^2\right\}$$

$$= \frac{15}{48}mv_0^2$$

3. Ans (9)

Resistance of bulb, $R_{bulb} = \frac{V_{bulb}^2}{P_{bulb}} = 0.5\Omega$

In order to the bulb glows at full intensity, current will pass through the bulb =

$$\frac{4.5W}{1.5V} = 3A$$

equivalent resistance of circuit, R_{eq} =

$$\frac{5}{3} + \frac{0.5\times 1}{0.5+1} = \frac{5}{3} + \frac{0.5}{1.5} = \frac{5}{3} + \frac{1}{3} = 2\Omega$$

$\therefore$ Current supplied by battery $\frac{E}{R_{eq}} = \frac{E}{2}$

Current in bulb = $I \times \frac{1}{(1+0.5)} \Rightarrow 3A =$

$$\frac{I}{1.5} = \frac{E}{2\times 1.5} \Rightarrow E = 9 \text{ volt.}$$

4. Ans (5)

Limiting friction between A & B = 90 N

Limiting friction between B & C = 80 N

Limiting friction between C & ground = 60 N

Since limiting friction is least between C and ground, slipping will occur at first between C and ground. This will occur when F = 60 N.

5. Ans (2)

$$T = 2\pi\sqrt{\frac{I}{C}} \Rightarrow$$

$$\frac{T'}{T} = \sqrt{\frac{I'}{I}} = \sqrt{\frac{\left(\pi\frac{4r^2}{2}\frac{t}{4}\right)4r^2}{\frac{(\rho\pi r^2 t)r^2}{2}}}$$

$T' = 2T$

CHEMISTRY

SEC-1

3. Ans (A,B,C,D)

Borax on heating produces H_2O (neutral oxide) + $NaBO_2$ + $B2O_3$.

4. Ans (A,C,D)

Decompoition reactions are endothermic in nature

$$\Delta_r S = \frac{\Delta_r H}{T} \text{ at equilibrium.}$$

$$\Delta_r G = \underbrace{\Delta_r H}_{+ve} - \underbrace{T\Delta_r S}_{-ve} \Rightarrow$$

(–ve) at high temperature.

5. Ans (A,B,D)

$$Ag(s) + Cl^- + H^+ \rightarrow AgCl(s) + \frac{1}{2}H_2$$

$$E = 0 - (0.18) - 0.06\log\frac{1}{[H^+][Cl^-]}$$

$= -0.18 - 0.06\times 6 = -0.54V$

$\therefore$ Cell will be represented as : Pt | H_2 | HCl | AgCl | Ag & Cell potential = 0.54 V

$$E^\circ_{Cl^-|AgCl|Ag} = E^\circ_{Ag^+|Ag} + 0.06\log K_{sp}(AgCl)$$

$$0.18 = E^\circ_{Ag^+|Ag} - 0.06\times 10$$

$$\Rightarrow E^\circ_{Ag^+|Ag} = 0.78 \text{ volt}$$

CHEMISTRY

SEC-2

1. Ans (D)

0.1 M HCOOK

$\Delta T_f = iK_f m \simeq 2 \times 0.1 \times K_f = 0.2\ K_f$

$\pi \simeq 0.2\ RT$

$$pOH = \frac{1}{2}[pK_w - pK_a - \log C] = \frac{1}{2}[14 - 4 + 1] = 5.5$$

pH = 8.5

Electrolysis products :

CO_2, H_2 at anode

: H_2 at cathode

2. Ans (B)

0.1 m $Ca(NO_3)_2$

$\Delta Tb = iK_b m = 0.3\ K_b$

$\pi \simeq 0.3\ RT$

Electrolysis products : H_2 at cathode : O_2 at anode

pH = 7 (salt of S.A + S.B)

3. Ans (A)

0.1 M CH_3COONa + 0.2 M $CH_3COOH \Rightarrow$

Acidic buffer solution

$$\pi = (i_1 M_1 + i_2 M_2)RT \simeq (0.1 \times 2 + 0.2)RT$$

$\simeq 0.4RT$

Electrolysis products : H_2 at cathode

: CO_2, C_2H_6 at anode

mainly

$$pH = pK_a + \log \frac{0.1}{0.2} = 5 - 0.3 = 4.7$$

CHEMISTRY

SEC-3

1. Ans (5)

$B = CO_2$

$D = C_2H_2$

2. Ans (4)

(iii), (iv), (v), (vii)

3. Ans (0)

x = 3 (K, Rb, Cs)

y = 2 (Mg, Be)

u = 5 (Concentrated solution of all alkali metals in liquid ammonia is blue coloured solution)

5. Ans (7)

$$\underset{\text{(l–form)}}{B} \underset{k_b}{\overset{k_f}{\rightleftharpoons}} \underset{\text{(d–form)}}{A}$$

t=0	a	0
t	a–x	x
$t = t_{eq}$	$(a–x_{eq})$	x_{eq}

at equilibrium $\frac{dx}{dt} = k_f(a–x_{eq}) – k_b x_{eq} = 0$

$$\Rightarrow K_{eq} = \frac{k_f}{k_b} = \frac{x_{eq}}{a - x_{eq}}$$

Where, $x_{eq} = \frac{a}{2} \Rightarrow k_f = k_b$

$$(k_f + k_b) = \frac{2.303}{t} \log \left(\frac{x_{eq}}{x_{cq} - x}\right)$$

$$\Rightarrow 2k_f = \frac{2.303}{230.3} \log \frac{a/2}{a/2 - a/10} = \frac{1}{100} \log \frac{0.5}{0.4}$$

$$= \frac{1}{100}(0.7 - 0.6) = 10^{-3} s^{-1}$$

$\therefore k_f = 5 \times 10^{-4}\ s^{-1} = k_b$

$\therefore k_{eq} = 1\ \ \Delta_r G^o = 0,\ \Delta_r G = –ve$

As A & B are enantiomers

$\Rightarrow (S_m^\circ)_A = (S_m^\circ)_B$

and $\Delta_r H° = 0,\ \Delta_r H = 0$

MATHEMATICS

SEC-1

1. Ans (B,D)

$\vec{a} \times \vec{b} = x\vec{a} + y\vec{b} + z\vec{c}$. Let $\vec{a} = \lambda \vec{b} \times \vec{c}$. $\lambda = \frac{1}{2}$

So, $\frac{\vec{b} \times \vec{c}}{2} \times \vec{b} = \frac{x}{2}\vec{b} \times \vec{c} + y\vec{b} + z\vec{c}$

$$\Rightarrow -\frac{\vec{b}}{2} + \frac{\vec{c}}{6} = \frac{x}{2}\left(\vec{b} \times \vec{c}\right) + y\vec{b} + z\vec{c}$$

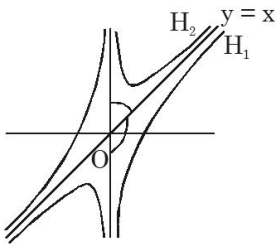
$$\Rightarrow x = 0,\ y = -\frac{1}{2},\ z = \frac{1}{6}$$

2. Ans (B,D)

Integrating we get ; $x^2 + (f(x))^2 + (f'(x))^2 = k$

so, f(x) and f '(x) are bounded

3. Ans (A,D)



asymptote of $x^2 = xy + 1$ and $x^2 + 1 = xy$

are $x^2 - xy = 0$

and $y^2 - xy - 1 = 0$ and $x^2 - xy + 1 = 0$ are conjugate of each other

$\Rightarrow \frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$ (direct result)

for $x^2 + 1 = xy$ if we rotate the conic by $\frac{\pi}{8}$ it will be of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

with asymptotes $y = \pm\frac{b}{a}x \Rightarrow \frac{b}{a} = \tan\frac{\pi}{8}$

$\Rightarrow e_2^2 = 1 + \frac{b^2}{a^2} = \frac{1}{1+\cos\frac{\pi}{8}}$

$\Rightarrow \frac{1}{e_1^2} = 1 - \frac{1}{e_2^2} = 1 - \frac{1}{\sec\frac{\pi}{8}} = 1 - \cos\frac{\pi}{8}$

$\left.\begin{array}{l} e_1^2 = \frac{1}{1-\cos\frac{\pi}{8}} \\ \Rightarrow \\ e_2^2 = \frac{1}{1+\cos\frac{\pi}{8}} \end{array}\right\} \Rightarrow e_1 > e_2$

4. Ans (D)

$p = \frac{2^{2\times2}}{2^{3\times3}} = \frac{1}{32}$

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Total number of matrices = 2^9

$(s_1s_2s_3t_1t_2t_3)$ is odd integer if

(1) all the elements are 1 (no of matrices = 1)

(2) four zeros and five ones (no of matrices = 9)

(zeros must occupy all the positions of any minor)

(3) six zeroes and three ones (no of matrices = 6)

$({}^3C_1.{}^2C_1.1)$

$\Rightarrow$ required probability $= \frac{16}{2^9} = \frac{2^y}{2^9}$

5. Ans (A,B,C,D)

Given $AA^T = I$ and $|A| = 1$

(A) $|A - I| = |A - AA^T| = |I - A^T|$

$= |(I - A^T)^T|$

$|A - I| = |I - A| = -|A - I|$

$\Rightarrow |A - I| = 0$

(B) $|A^2 - I| = |A^2 - AA^T|$

$= |A|\,|A - A^T| = |A - A^T|$

since $A - A^T$ is skew symmetric

$\Rightarrow |A - A^T| = 0 \Rightarrow$ hence is not invertible

(C) $|A - A^T| = 0 \Rightarrow (A - A^T)^2$ is not invertible

(D) $(A - (\det A)) = A - I$ is not invertible

6. Ans (B,C,D)

Replace x by –x we get ;

$(2x^2 + 2x + 1)^n = \sum_{r=0}^{2n} a_r x^r$

So, $\sum_{r=0}^{2n} a_r = 5^n, \sum_{r=0}^{2n} (-1)^r a_r = 1$. Also,$a_0 = 1$

again,

$\sum_{r=0}^{2n} a_r x^r = \left((x+1)^2 + x^2\right)^n = \sum {}^n c_k x^{2k} (x+1)^{2n-2k}$

$= \sum \left[{}^n c_k \sum {}^{2n-2k} c_t x^{2n-t} \right]$

⇒ comparing coeff. of x^{2n-r}, we get

$\sum {}^n c_k {}^{2n-2k} C_r = a_{2n-r} = 2^{n-r} a_r$

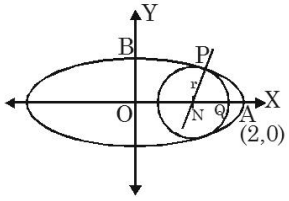
Also, $\sum_{r=0}^{2n} (-1)^r . a_r a_{2n-r}$ = coeff. of x^{2n}

in $\left((2x^2 - 2x + 1)^n (2x^2 + 2x + 1)^n\right)$

= coeff. of x^{2n} in $(4x^4 + 1)^n = 0$ if n odd.

7. Ans (B,C)

$\frac{x^2}{4} + \frac{y^2}{3} = 1$



N ≡ (k, 0) & Q(r + k, 0)

Normal PN at

$P(2\cos\theta,\ \sqrt{3}\sin\theta)\ ;\ \theta \neq \frac{n\pi}{2}$ is

$2x\sec\theta - \sqrt{3}y\operatorname{cosec}\theta = 1$

$\Rightarrow N \equiv \left(\frac{\cos\theta}{2},\ 0\right) \Rightarrow$

$k = \frac{\cos\theta}{2} \in \left(0,\ \frac{1}{2}\right)$

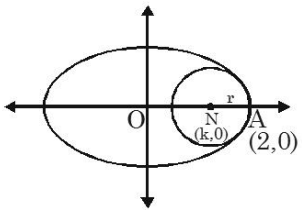
[as $k > 0$ & $\theta \neq \frac{n\pi}{2}$]

So, all such circles touches the arc of the ellipse in the Ist (as well as 4th) quadrant between (strictly) A & B.

Thus, $0 < k < \frac{1}{2}$ & $r + k < 2$ (as $Q_x < A_x$)

Now, these is another possibility when point of normalcy is A (note it cannot be B otherwise k = 0)

In this case,



$r + k = 2$

So, circle is

$(x - k)^2 + y^2 = (2 - k)^2$

Intersecting it with given ellipse :

$3x^2 + 4[(2 - k)^2 - (x - k)^2] = 12$

$\Rightarrow x^2 - 8kx + 16k - 4 = 0$

$\Rightarrow x = 2, 8k - 2$

Since, the circle lies completely inside the ellipse So, other point of intersection

(i.e. $x = 8k - 2$) should be imaginary.

Hence, $8k - 2 > 2$ (Note $8k - 2 > -2$ as $k > 0$)

$\Rightarrow k >$ & $r + k = 2$

MATHEMATICS

SEC-2

1. Ans (C)

$x + y = 1$ represents a line

$x^2 + y^2 = 1$ represents a circle one of whose tangents is $x + y = \sqrt{2}$ at point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

where as, $\sqrt{x} + \sqrt{y} = 1 \Rightarrow y = 1 + x - 2\sqrt{x}$

$\Rightarrow 4x = (1 + x - y)^2$ represents a parabola.

and $\frac{1}{x} + \frac{1}{y} = 1$ represents a hyperbola.

2. Ans (C)

$x + y = 1$ represents a line

$x^2 + y^2 = 1$ represents a circle one of whose tangents is $x + y = \sqrt{2}$ at point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

where as, $\sqrt{x} + \sqrt{y} = 1 \Rightarrow y = 1 + x - 2\sqrt{x}$

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3. Ans (D)

$x + y = 1$ represents a line

$x^2 + y^2 = 1$ represents a circle one of whose tangents is $x + y = \sqrt{2}$ at point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

where as, $\sqrt{x} + \sqrt{y} = 1 \Rightarrow y = 1 + x - 2\sqrt{x}$

$\Rightarrow 4x = (1 + x - y)^2$ represents a parabola.

and $\frac{1}{x} + \frac{1}{y} = 1$ represents a hyperbola.

4. Ans (B)

(P) $[x]\{x\} = 1 \Rightarrow x \in \left\{n + \frac{1}{n} : n \in I,\ n \geq 2\right\}$

(Q) $[x]\{x\} = 1 \Rightarrow x \in \left\{n + \frac{1}{|n|} : n \in I,\ |n| \geq 2\right\}$

(R) $\left[x\left[\frac{1}{x}\right]\right] = 1 \Rightarrow x \in (-2, 0) \cup \left\{\frac{1}{n} : n \in I^+\right\}$

(S) $\{x\} + \left\{\frac{1}{x}\right\} = 1 \Rightarrow x \in \left\{\frac{n \pm \sqrt{n^2 - 4}}{2} : n \in I,\ |n| \geq 3\right\}$

5. Ans (D)

(P) $[x]\{x\} = 1 \Rightarrow x \in \left\{n + \frac{1}{n} : n \in I,\ n \geq 2\right\}$

(Q) $[x]\{x\} = 1 \Rightarrow x \in \left\{n + \frac{1}{|n|} : n \in I,\ |n| \geq 2\right\}$

(R) $\left[x\left[\frac{1}{x}\right]\right] = 1 \Rightarrow x \in (-2, 0) \cup \left\{\frac{1}{n} : n \in I^+\right\}$

(S) $\{x\} + \left\{\frac{1}{x}\right\} = 1 \Rightarrow x \in \left\{\frac{n \pm \sqrt{n^2 - 4}}{2} : n \in I,\ |n| \geq 3\right\}$

6. Ans (C)

(P) $[x]\{x\} = 1 \Rightarrow x \in \left\{n + \frac{1}{n} : n \in I,\ n \geq 2\right\}$

(Q) $[x]\{x\} = 1 \Rightarrow x \in \left\{n + \frac{1}{|n|} : n \in I,\ |n| \geq 2\right\}$

(R) $\left[x\left[\frac{1}{x}\right]\right] = 1 \Rightarrow x \in (-2, 0) \cup \left\{\frac{1}{n} : n \in I^+\right\}$

(S) $\{x\} + \left\{\frac{1}{x}\right\} = 1 \Rightarrow x \in \left\{\frac{n \pm \sqrt{n^2 - 4}}{2} : n \in I,\ |n| \geq 3\right\}$

MATHEMATICS

SEC-3

1. Ans (7)

at $A = \frac{\pi}{7}, B = \frac{2\pi}{7}, C = \frac{4\pi}{7}$ which are angles of a triangle.

So, x = tanA, y = tanB, z = –tanC.

$$\frac{1}{2}(x^2 + y^2 + z^2 - x^2y^2z)$$

$$= \frac{1}{2}\left[\tan^2 A + \tan^2 B + \tan^2 C - (\tan A \tan B \tan C)^2\right]$$

$$= -(\tan A \tan B + \tan B \tan C + \tan C \tan A)$$

$$= -\frac{\sin A \sin B \cos C + \sin B \sin C \cos A + \sin C \sin A \cos B}{\cos A \cos B \cos C}$$

$$= \frac{\cos(A + B + C) - \cos A \cos B \cos C}{\cos A \cos B \cos C}$$

$$= -\frac{1}{\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}} - 1 = 8 - 1 = 7.$$

2. Ans (6)

$f'(x) = 0 \Rightarrow x = -\frac{9}{5a}, \frac{1}{a}$

As, a < 0 and maxima comes first so,

$$\frac{1}{a} = -\frac{5}{9} \Rightarrow a = -\frac{9}{5} \text{ and } \alpha = 1$$

Now, $f(1) > 0 \Rightarrow b > \frac{36}{5} \Rightarrow a + b > \frac{27}{5}$

3. Ans (5)

$$\text{Let } t = \frac{x^3 - 3x + 1}{x^2 - x} = \frac{(x^2 - x)(x + 1) - (x - 1) - x}{x^2 - x}$$

$$= x + 1 - \frac{1}{x} - \frac{1}{x - 1}$$

So, I =

$$\int_{-\frac{17}{12}}^{-\frac{1}{6}} \frac{1}{t^2}dt = \left(-\frac{1}{t}\right)\Bigg|_{-17/12}^{-1/6} = 6 - \frac{12}{17} = \frac{90}{17}$$

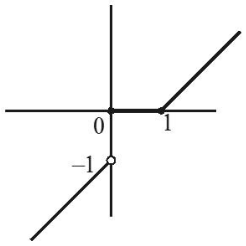
(for x = –2, $t = -\frac{1}{6}$ and for x = –3, $t = -\frac{17}{12}$)

4. Ans (4)

g is symmetrical about

$$x = \frac{\pi}{2}.\ g_{max} = g\left(\frac{\pi}{2}\right) = 2f\left(\frac{\pi}{2}\right) = 4$$

5. Ans (2)



Not differentiable at x = 0, 1

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
12-06-2021

JEE(Main + Advance) : JEE (MAIN+ADVANCED) NURTURE PHASE-I & II_MT#03 (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 12-06-2021

ANSWER KEY PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	A,B,D	B,C,D	A,B,D	B,C,D	A,B	A,B
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	A	D	D	A	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	2	9	5	2		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	A,B,C,D	A,B,C,D	A,C,D	A,B,D	A,B,C	A,B,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	D	B	A	D	B	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	4	0	8	7		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	B,D	A,D	D	A,B,C,D	B,C,D	B,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	C	D	B	D	C	
SEC-3	Q.	1	2	3	4	5		
	A.	7	6	5	4	2		

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE (Advanced)
MAJOR
12-06-2021

JEE(Main + Advance) : IIT-JEE(MAIN + ADVANCED) ENTHUSIAST & LEADER (PHASE : ALL PHASES)

Test Type : MAJOR Test Pattern : JEE (Advanced)

TEST DATE : 12-06-2021

ANSWER KEY PAPER-2

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B	A	A	A	A	D	A
SEC-2	Q.	1	2	3	4	5	6	7
	A.	B,C	A,B,C	A,B,C	A,B,C	A,B,C	B,C	A,C
SEC-3	Q.	1	2	3	4			
	A.	B	B	A	A			

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B	D	C	C	D	B	A
SEC-2	Q.	1	2	3	4	5	6	7
	A.	C,D	A,D	B,D	A,B,C,D	B	A,B,C,D	A,C,D
SEC-3	Q.	1	2	3	4			
	A.	C	C	A	B			

MATHEMATICS

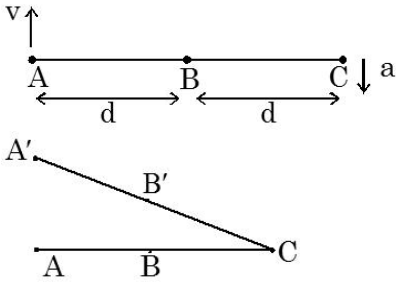
SEC-1	Q.	1	2	3	4	5	6	7
	A.	A	A	D	A	C	A	B
SEC-2	Q.	1	2	3	4	5	6	7
	A.	A,B	B,C,D	A,B	A,B,C	A,B,C	C,D	A
SEC-3	Q.	1	2	3	4			
	A.	C	C	A	B			

HINT – SHEET

PHYSICS

SEC-1

2. Ans (A)



Let us observe the motion of A and B relative to C.

$$\frac{AA'}{2d} = \frac{BB'}{d}$$

$$BB' = \frac{AA'}{2} = \frac{vt + \frac{1}{2}at^2}{2}$$

$$BB' = \frac{Vt}{2} + \frac{1}{4}at^2$$

$\frac{V}{2}$ is initially speed of B w.r.t. C as well as ground.

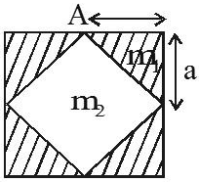
$\frac{a}{2}$ is acceleration of B w.r.t. C

$$\vec{a}_{B/g} = \vec{a}_{B/C} + \vec{a}_{C/g}$$

$$= \frac{a}{2} - a = \frac{-1}{2}a$$

4. Ans (A)

$$I_1 = \frac{I_A}{2}$$



$$m = (\sigma a^2)$$

$$m_1 = \sigma(2a)^2 = 4m$$

$$m_2 = \sigma(\sqrt{2}a)^2 = 2m$$

$$I_A = \left(\frac{m_1(2a)^2 + m_1a^2}{6}\right) - \left(\frac{m_2(\sqrt{2}a)^2 + m_2a^2}{6}\right)$$

$$I_A = 4ma^2\left(\frac{2+1}{3}\right) - 2ma^2\left(\frac{1}{3}+1\right)$$

$$= ma^2\left[\frac{20-8}{3}\right] = 4ma^2$$



$$I_1 = 2ma^2$$

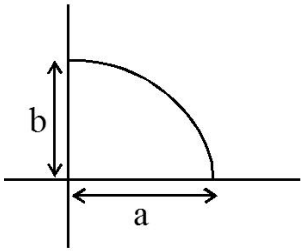
$$I_2 = \frac{ma^2}{6} + m\left(\frac{a}{\sqrt{2}}\right)^2$$

$$I_2 = ma^2\left(\frac{1}{6}+\frac{1}{2}\right) \Rightarrow \frac{ma^2}{2}\ \frac{4}{3}$$

$$I_2 = \frac{2ma^2}{3}$$

$$\frac{I_1}{I_2} = 3$$

5. Ans (A)



$$x = \frac{\int dm\, x}{\int dm} = \frac{\int dv\, x}{\int dv}$$

$$\int dv = \int \pi y^2 dx;\ \frac{y^2}{b^2} = \frac{1-x^2}{a^2}$$

$$= \pi b^2 \int \left(\frac{1-x^2}{a^2}\right) dx$$

$$= \pi b^2 \left[\frac{x - x^3}{3a^2}\right]_0^a$$

$$= \pi b^2 \left(\frac{a-a}{3}\right) = \frac{2\pi}{3}b^2a$$

$$\int dvx = \int \pi y^2\, dx\, x$$

$$= \pi b^2 \int \left(1 - \frac{x^2}{a^2}\right) x dx$$

$$= \pi b^2 \left(\frac{x^2}{2} - \frac{x^4}{4a^2}\right)_0^a$$

$$= \pi b^2 \left(\frac{a^2}{2} - \frac{a^4}{4}\right) = \mu b^2 \frac{a^2}{4}$$

$$\left(X_{cm} = \frac{\frac{\pi b^2 a^2}{4}}{\frac{2\pi}{3}b^2a} = \frac{3}{8}a\right)$$

6. Ans (D)

$mv = -mv_1 + Mv_2$

$t = \frac{d}{v} + \frac{d}{ev} + \frac{d}{e^2 v} + \ldots\ldots\ldots$

$t = \frac{d}{v}\left(1 + \frac{1}{e} + \frac{1}{e^2} + \ldots\ldots\right)$

Common ratio > 1 so $t \to \infty$

7. Ans (A)

$\sqrt{f} = (Z - b)\left(1 - \frac{1}{n^2}\right)^{1/2}$

PHYSICS

SEC-2

1. Ans (B,C)



$R_A = 100\Omega$

$I_f = 50 \times 10^{-6}$

(a) $V_{max} = 50 \times 10^{-6}[100 + 10 \times 10^3]$

$V_{max} \cong 50 \times 10^{-6} \times 10^4 \cong 0.5V$

(b) $V_{max} = 50 \times 10^{-6}[100 + 200 \times 10^3]$

$\cong 50 \times 10^{-6} \times 200 \times 10^3 \cong 10V$

(c) $50 \times 10^{-6} \times 100 = x\,5 \times 10^{-3}$

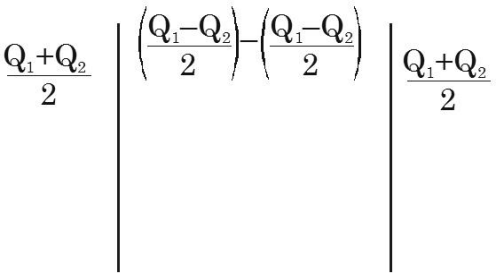
$x = 10A \qquad I_{max} \cong 10A$

(d) $50 \times 10^{-6} \times 100 = x$

$x = 5 \times 10^{-6}A \quad I_{max} = 55 \times 10^{-6}A$

2. Ans (A,B,C)

Charges on the inner and outer plates will be



$q = \left(\frac{\theta_1 - \theta_2}{2}\right) e^{-t/RC}$

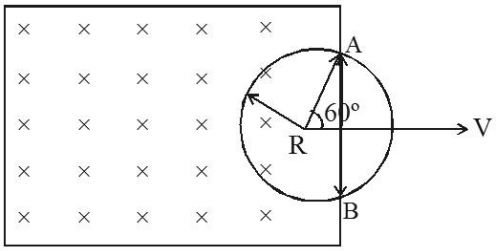
3. Ans (A,B,C)

At any time t

$$I = \frac{\varepsilon}{\lambda 2\pi R} = \frac{B(2AB)v}{\lambda 2\pi R}$$

$$I = \frac{Bv}{\lambda \pi R}\sqrt{R^2 - \left(\frac{R}{2} - vt\right)^2}$$

$$I = \frac{Bv}{\lambda \pi}\sqrt{1 - \left(\frac{1}{2} - \frac{vt}{R}\right)^2}$$

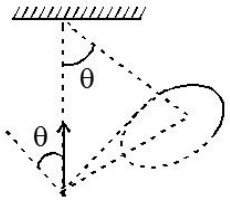


$$I = \frac{2Bv}{\lambda 2\pi R}\sqrt{R^2 - \frac{R^2}{4} + Rvt - v^2t^2}$$

$$I = \frac{Bv}{\lambda 2\pi R}\sqrt{3R^2 + 4Rvt - 4v^2t^2}$$

5. Ans (A,B,C)

$$g_{eff} = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2}$$



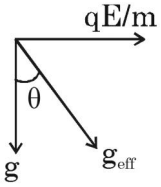
$T\cos\theta = m\, g_{eff}$

$T\cos\theta = mR\,\omega^2$

$$\frac{R\omega^2}{g_{eff}} = \tan\theta$$

$$\omega^2 = g_{eff}\frac{\tan\theta}{\ell\sin\theta} \Rightarrow \frac{g_{eff}}{\ell\cos\theta}$$

$$\omega = \sqrt{\frac{g_{eff}}{\ell\cos\theta}}$$



$$\cos\theta = \frac{I}{g_{eff}}$$

$$\sin\theta = \frac{qE}{mg_{eff}}$$

Time period

$$= \frac{2\pi}{\omega} = \frac{2\pi}{g_{eff}}\sqrt{\ell g} = 2\pi\sqrt{\frac{\ell g}{g^2 + \left(\frac{qE}{m}\right)^2}}$$

$$v = Rw = \ell\sin\theta\sqrt{\frac{g_{eff}}{\ell\cos\theta}}$$

$$v = \ell\frac{qE}{mg_{eff}}\cdot\sqrt{\frac{g^2_{eff}}{\ell g}}$$

$$v = \frac{qE}{m}\cdot\sqrt{\frac{\ell}{g}}$$

$$T = \frac{m\,g^2_{eff}}{g} \Rightarrow \frac{m}{g}\left[g^2 + \left(\frac{qE}{m}\right)^2\right]$$

$$T = \left[mg + \frac{q^2E^2}{mg}\right]$$

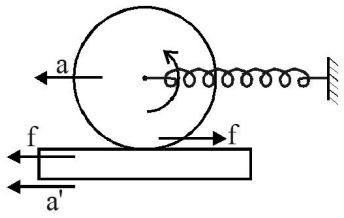
6. Ans (B,C)

$kx - f = Ma$

$f = Ma'$

$a' = a - R\alpha$

$fR = \frac{1}{2} MR^2 a$



$\frac{f}{M} = a - \frac{2f}{M}$

$\frac{3f}{M} = a$

$a' = \frac{3f}{M} - \frac{2f}{M} = \frac{f}{M}$

7. Ans (A,C)

Intensity is maximum when the area over which the power is distributed is minimum i.e. at the point 1.

Frequency is max. when relative velocity of approach is maximum.

PHYSICS

SEC-3

1. Ans (B)

$\Delta x = \mu_1 SS_2 + \mu_2 S_2 P$

$- \left[\mu_1 SS_1 + \mu_2 (S_1 P - t) + \int \mu_3 dx\right]$

from graph $\int \mu_3 dx = \text{Area}$

$= \frac{1}{2}(1+3)t \Rightarrow 2t$

$\Delta x = \mu_1 SS_2 + \mu_2 S_2 P$

$- [\mu_1 SS_1 + \mu_2 S_1 P - \mu_2 t + 2t]$

In order to get central maxima at centre

$\Delta x = 0$

$\Delta x =$

$-\mu_1 [SS_2 - SS_1] + \mu_2 [S_2 P - S_1 P] + \mu_2 t - 2t$

$SS_2 - SS_1 = \sqrt{D^2 + d^2} - D$

$\Rightarrow D\left[1 + \frac{d^2}{D^2}\right]^{1/2} - D$

$= D\left[1 + \frac{1}{2}\frac{d^2}{D^2}\right] - D$

$SS_2 - SS_1 = \frac{1}{2}\frac{d^2}{D}$

$\Delta x = \mu_1 \left(\frac{1}{2}\frac{d^2}{D}\right) + \mu(0) + \mu_2 t - 2t$

$2t - \mu_2 t = 2 \times \frac{1}{2}\frac{(10^{-3})^2}{(1m)}$

$2t - \frac{3t}{2} = 10^{-6}$

$\frac{t}{2} = 10^{-6}$

$t = 2 \times 10^{-6}$ m ($t = 2\mu m$)

2. Ans (B)

$\Delta x = \mu_1 [SS_2 - SS_1] + \mu_2 [S_2P - S_1P] + \mu_2 t - 2t$

For central maxima

$$0 = 2\frac{1}{2}\frac{d^2}{D} + \frac{3}{2}(S_2P - S_1P) + \frac{3t}{2} - 2t$$

$$\frac{3}{2}(S_2P - S_1P) = 2t - \frac{3t}{2} - \frac{d^2}{D}$$

$$(S_2P - S_1P) = \frac{2}{3}\left[\frac{t}{2} - 10^{-6}\right]$$

$$\frac{2}{3}\left[\frac{10^{-6}}{2} - 10^{-6}\right]$$

$$S_2P - S_1P \Rightarrow -\frac{1}{3}\mu m$$

$$d\sin\theta \cong d\tan\theta$$

$$d\frac{y}{D} = -\frac{1}{3}\mu m$$

$$y = -\frac{1}{3}\frac{10^{-6}\times 1}{10^{-3}} m$$

$$y = -\frac{1}{3} mm$$

3. Ans (A)

$$\eta_{th} = \frac{\text{Net workdone}}{\text{Net heat added}}$$

Since processes 1-2 and 3-4 are adiabatic processes, the heat transfer during the cycle takes place only during processes 2-3 and 4-1 respectively. Therefore, thermal efficiency can be written as,

$$\eta_{th} = \frac{\text{Heat added} - \text{Heat rejected}}{\text{Heat added}}$$

Consider 'm' kg of working fluid,

Heat added = $mC_V (T_3 - T_2)$

Heat Rajected = $mC_V (T_4 - T_1)$

$$\eta_{th} = \frac{mC_V(T_3 - T_2) - mC_V(T_4 - T_1)}{mC_V(T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

For the reversible adiabatic processes 3-4 and 1-2, we can write,

$$\frac{T_4}{T_3} = \left(\frac{v_3}{v_4}\right)^{\gamma-1} \text{ and } \frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{\gamma-1}$$

$v_2 = v_3$ and $v_4 = v_1$

$$\frac{T_4}{T_3} = \frac{T_1}{T_2} = \frac{T_4 - T_1}{T_3 - T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\eta_{th} = 1 - \frac{T_1}{T_2} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

The ratio $\frac{V_1}{V_2}$ is called as compression ratio, r.

$$\eta_{th} = 1 - \left(\frac{1}{r}\right)^{\gamma-1}$$

Net work done = $mC_V\{(T_3 - T_2) - (T_4 - T_1)\}$

Displacement volume = $(V_1 - V_2)$

$$= V_1\left(1 - \frac{1}{r}\right) = \frac{mRT_1}{P_1}\left(\frac{r-1}{r}\right)$$

$$= \frac{mC_V(\gamma-1)T_1}{P_1}\left\{\frac{r-1}{r}\right\}$$

Since, $R = C_V(\gamma - 1)$

4. Ans (A)

$$\text{mep} = \frac{mC_V[(T_3 - T_2) - (T_4 - T_1)}{\frac{mC_V(\gamma-1)T_1}{P_1}\left\{\left(\frac{r-1}{r}\right)\right\}}$$

$$= \left(\frac{1}{\gamma-1}\right)\left(\frac{P_1}{T_1}\right)\left(\frac{r}{r-1}\right)\{(T_3 - T_2) - (T_4 - T_1)\}$$

CHEMISTRY

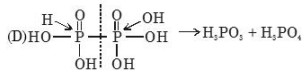
SEC-2

3. Ans (B,D)

(A) $(Mg(OH)_2 + H_3C - C \equiv CH)$ are produced through a non-redox reaction

(B) N of NH_3 does not have vacant orbital.

(C) $(H_3BO_3 + NH_3 + H_2)$ are produced when $B_2N_2H_6$ react with water on heating.

(D)  $\rightarrow H_3PO_3 + H_3PO_4$

4. Ans (A,B,C,D)

(A) CO_2 is produced

(B) SO_2 is produced

(C) CO & CO_2 are produced

(D) NO_2 and O_2 are produced.

MATHEMATICS

SEC-1

1. Ans (A)

Let $(\lambda, \lambda, \lambda)$ point of intersection then $\lambda(\sin A + \sin B + \sin C) = 2d^2$ and

$\lambda(\sin 2A + \sin 2B + \sin 2C) = d^2$

dividing $\dfrac{\sum \sin 2A}{\sum \sin A} = \dfrac{1}{2}$

$\Rightarrow \sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2} = \dfrac{1}{16}$

2. Ans (A)

$(1 + y^2e^{-2x})dy = y^3e^{-2x}dx$

$\dfrac{dy}{y} + ye^{-2x}dy - y^2e^{-2x}dx = 0$

Now integrate

3. Ans (D)

$g(x) = f(x^2 - 2x - 1) + f(5 - x^2 + 2x)$

$= 2x^4 - 8x^3 - 4x^2 + 24x + 18$

$g'(x) = 8x^3 - 24x^2 - 8x + 24 = 0 \Rightarrow x = -1, 1, 3$

$g(x) \geq \min\{g(-1), g(1), g(3)\} = 0$

$g(x) \geq 0 \ \forall \ x \in R$

4. Ans (A)

$\bar{d} = x\bar{a} + y\bar{b} + z\bar{c}$ take dot product with

$\bar{b} \times \bar{c}, \ \bar{c} \times \bar{a}, \ \bar{a} \times \bar{b}$

$$\bar{d} = \frac{[\bar{d}\ \bar{b}\ \bar{c}]\bar{a} + [\bar{d}\ \bar{c}\ \bar{a}]\bar{b} + [\bar{d}\ \bar{a}\ \bar{b}]\bar{c}}{[\bar{a}\ \bar{b}\ \bar{c}]}$$

5. Ans (C)

$(2a)^2 + b^2 = \sin^2 x$, $(2b)^2 + a^2 = \cos^2 x$ where 3a and 3b legs of triangle then hypotenuse

$= 3\sqrt{a^2 + b^2} = \frac{3\sqrt{5}}{5}$

6. Ans (A)

Equation of directrix is $x - 2y = 0$

7. Ans (B)

$A^2 = I$ then $\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}^2 = I$

$\Rightarrow a = -1$ hence $|A| = -1$

$A^{-1} = A \Rightarrow |2A| = -8$ hence $|3\text{adj}.2A|$

$= 27(-8)^2 = 1728$

MATHEMATICS

SEC-2

1. Ans (A,B)

Let circle $(x - h)^2 + (y - k)^2 = r^2$

$r = \left|\frac{ah + bk + 1}{\sqrt{a^2 + b^2}}\right| \Rightarrow a^2(h^2 - r^2) + b^2(k^2 - r^2) +$ $2abhk + 2ah + 2bk + 1 = 0$

comparing, we get centre (3, 0), $r = \sqrt{5}$

2. Ans (B,C,D)

AM > GM

$2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} > 3\left(2^{\frac{\pi}{2} + \tan^{-1}x}\right)^{1/3}$

$2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} > 3$

No solution.

3. Ans (A,B)

differentiable w.r.t. x,

we get

$$\frac{(mx\cos 4x - \ell \sin 4x)}{x^2} = \frac{\ell(4x\cos x - \sin 4x)}{x^2}$$

$m = 4\ell$

Put $m = 4\ell$ to get

$$\ell \int_0^x \left(\frac{4\cos 4t}{t} - \frac{\sin 4t}{t^2}\right) dt = \frac{\ell \sin 4x}{x} - 1$$

$$4\int_0^x \frac{\cos 4t}{t} dt + \int_0^x \sin 4t \left(-\frac{1}{t^2}\right) dt = \frac{\sin 4x}{x} - \frac{1}{\ell}$$

By parts $\frac{\sin 4x}{x} - 4 = \frac{\sin 4x}{x} - \frac{1}{\ell} \Rightarrow \ell = \frac{1}{4}$

$m = 1$

4. Ans (A,B,C)

Any tangent $y = mx \pm 4\sin\theta\sqrt{-m}$ it will pass through centre of circle if (1, 1) satisfy this

$m^2 + 2\,[8\sin^2\theta - 1]\,m + 1 = 0$

$D < 0 \Rightarrow \sin^2\theta < \frac{1}{4}$

$\theta \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, \frac{7\pi}{6}\right) \cup \left(\frac{11\pi}{6}, 2\pi\right)$

6. Ans (C,D)

$$A = \left|2\int_0^1 \sqrt{1-x}\,dx\right| + \left|2\int_{-1}^0 \sqrt{x+1}\,dx\right| - (\sqrt{2})^2 = \frac{2}{3}$$

$$B = 2\left(\frac{1}{2}\times 1 - \int_0^1 x^2 dx\right) = \frac{1}{3}$$

7. Ans (A)

$$x^2 - \frac{2p}{p-5}x + \frac{p-4}{p-5} = 0$$

$f(0) > 0$, $f(2) < 0$, $f(3) > 0$

MATHEMATICS

SEC-3

1. Ans (C)

No. of subsets having any of first $k-1$ elements $= 2^{k-1}$

2. Ans (C)

No. = total no. – subsets having even integers

3. Ans (A)

Total points generated = 20, two selected = $^{20}C_2$

No. of points with all odd co-ordinates = 4

No. of points with x odd, y even = 6

No. of points with x even, y odd = 4

No. of points with both even = 6

required number of points

$= {}^4C_2 + {}^6C_2 + {}^4C_2 + {}^6C_2 = 42$

$$P = \frac{42}{190}$$

4. Ans (B)

If S is the reduced sample space then $n(S) = {}^5C_2 \times 4 = 40$; as there are four vertical lines. On each line we need to choose points such that their mid-point belong to set S only. So, we choose both y-coordinates odd or even.

Since, $y \in \{0, 1, 2, 3, 4\}$

So, $n(E) = 4 \times (1 + {}^3C_2) = 16$

$$\Rightarrow P(E) = \frac{16}{40} = \frac{2}{5}$$